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Experiment-Driven Bayesian Optimization of the Impact Response of Multi-Material 3D-Printed Tensegrity-Inspired Structures

Tensegrity-inspired structures of rigid and soft members could cushion impacts, but their transmitted shock is hard to predict by simulation. We report a drop-test-driven Bayesian optimization campaign over three-strut prisms printed in one piece from polylactic acid (PLA) and thermoplastic polyurethane (TPU). The search space has five geometry variables, plus infill densities and filament settings from the third batch. We scale each candidate to a common mass target before printing, then 3D print and drop-test it repeatedly. We score designs by transmissibility (measured as a ratio of filtered top-to-base peak accelerations) and a mass-weighted rebound index (specimen mass times the squared fraction of impact velocity returned at the top), both minimized. Most specimens amplify the peak shock. Over a quasi-random initial batch and three model-recommended batches (35 specimens), the dominated hypervolume grew 56 percent, and the model chose every design on the final Pareto front. The final batch's best-predicted design lowered transmissibility to 0.803, 10 percent below the best initial-batch specimen. We printed one batch three times to measure repeatability. Print-to-print scatter in transmissibility was near 1 percent, and a seating check flagged both large outliers, so single-print extremes stay provisional until replicated. Code and data are available at https://github.com/vertical-cloud-lab/tensegrity-optimization.

Keywords: tensegrity, multi-material 3D printing, Bayesian optimization, impact mitigation, design of experiments

Draft note (remove before submission): this draft is written at the intended final scope of the study, and any value printed in this violet color is a synthetic placeholder rather than a measurement. The Results contain no violet values. The four reported batches (35 specimens) and the second and third prints of the batch-3 designs (18 specimens, the third added 2026-10-11) are measured data from the committed campaign record. The Conclusions carry one violet placeholder for the outcome of the replicate study listed below. Still pending, and written as future work: the re-seat confirmations of the batch-4 record specimen and of the third-print outlier dran35, the entry of the third print into the surrogate fit. Also pending are the drop tests of the replicate rank-stability study (27 specimens of three batch-4 designs, printed, with the plate-1 record withheld), the quasi-static crush campaign, the metal-analog comparison, and the archival (Zenodo) code release. Tables and figures moved from the main text for length are in the Supplementary Information, which the main text cites by S-number.

1 Introduction

Tensegrity structures are assemblies of rigid compression members suspended within a continuous network of tension members. Their defining trait is that no two rigid members contact one another, yielding lightweight assemblies with high strength-to-weight ratios and tunable nonlinear mechanical responses [1, 2]. These properties have made tensegrity attractive from metamaterial unit cells [3–5] to compliant impact-absorbing robots developed for planetary landing [6–10]. Recent work has shown that *tensegrity-inspired* unit cells can be directly 3D-printed and exhibit load-limiting plateaus under drop-weight impact [11]. Throughout this paper, we use *tensegrity-inspired* to refer to printable cells that borrow the geometry of traditional tensegrity structures without the defining mechanical conditions of pretensioned members, or cables, associated with classical tensegrity. In our specimens, the 3D printed cables are not tensioned as in a true tensegrity structure. Additionally, the flexible cables share a junction within the vertices of each rigid strut. Therefore, we do not claim our specimens to be classical tensegrity structures.

Multi-material fused-deposition modeling (FDM) can be used to pair a rigid polymer and a soft elastomer within the same part. It has been used to fabricate energy absorbers which consist of both soft and rigid portions, including thick-panel origami composites composed of polylactic acid (PLA) panels and thermoplastic polyurethane (TPU) hinges [12], as well as honeycombs of acrylonitrile butadiene styrene (ABS) interspersed in layers with TPU [13]. This rigid-soft multi-material printing capability makes it possible to fabricate diverse tensegrity-inspired geometries as well. Although extruded TPU does not behave as an ideal inextensible cable, its rate-dependent damping is a desirable property for impact-absorption applications.

Designing such structures capable of energy absorption is a *Design of Experiments* problem. Strut geometry, cable cross-section, connectivity, and unit-cell tiling span a large configuration space. Each candidate must be physically printed and tested. Its performance has to be estimated from repeated drops, because the scatter between drops and between prints can be as large as the differences between designs. The two relevant impact-performance metrics (the shock transmitted across the structure and the veloc-

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ity it gives back to the protected payload) also partly conflict, so a design that improves one can worsen the other. Bayesian optimization (BO) suits this regime. It chooses each next experiment from a probabilistic model fitted to every result so far, it accounts for measurement noise, and it can trade off two objectives. BO is widely used to tune machine-learning models, including AlphaGo [14], and increasingly to guide physical experiments in materials research, from sustainable concrete [15] to organic laser emitters [16]. Closest to this work, it has been applied to the design of architected materials [17, 18]. The gap this paper addresses is the combination of these two ideas: multi-material printing of tensegrity-inspired structures, and BO driven by physical experiments. Pajunen et al. [11] printed tensegrity-inspired cells under a single fabrication condition with no optimization loop, and Mo et al. [17] closed a multifidelity BO loop entirely in simulation. Among the studies reviewed here, prior work separately demonstrates physical impact testing of printed tensegrity-inspired structures, rigid-flexible additive fabrication, and Bayesian optimization of architected materials. We found no prior report combining these elements in a BO campaign over co-printed multi-material tensegrity-inspired hardware with physical impact measurements as the optimization ground truth. In the campaign reported here, we close the recommend, fabricate, measure, and update cycle three times. We trained the surrogate on a seed batch drawn from a Sobol sequence, and each of three successive model-recommended batches was fabricated, tested under the identical protocol, and folded back into the refit that produced the next.

Our motivating application is protecting the payload of a planetary lander, where added mass is costly. An absorber there should limit the peak shock delivered to the payload during touchdown and the motion it sends back into the payload afterward. These two needs set the two objectives of this paper. The transmissibility t_{180} compares the peak acceleration at the top of the structure with the peak at its base. The mass-weighted rebound index I_{reb} grows with the specimen's mass and with the share of the impact velocity it returns (Section 3.3). The mass budget sets a third condition, that we print every candidate at a common mass. The printed PLA/TPU T_3 prism studied here is a testbed for developing and validating the closed-loop workflow. It is not flight hardware, and it is not the end of the design space we intend to search. Our contribution is the loop itself, a stepping stone toward optimizing tensegrity-inspired structures generally, independent of the eventual application scale.

Contributions. This paper makes the following contributions:

(1) A parameterized family of multi-material 3D-printable tensegrity-inspired unit cells that uses TPU in place of the tension elements of the traditional tensegrity topology, with PLA members that carry compression most of the time. We do not prestress the cells, so neither member type acts as a pure tension or compression element. Under impact, some tendons stretch while others buckle. We anchor the TPU tendons *inside* the ends of each PLA strut, the strut acting as a rigid cage in which the tendons meeting at a given end join before exiting through discrete outlets. We report the resulting printable joint geometry. Pull-out strength and fatigue life remain untested and are an area of planned follow-on work. We compare candidate designs fairly through a constant-mass projection that scales every geometry to a common material budget before printing (Section 3.1).

(2) A characterized drop-tower protocol and two impact objectives for printed absorbers. We drop-test each specimen repeatedly (99 stabilized drops per specimen in the seed batch and 18 to 20 in later batches, after a 2-drop warm-up) with standardized SAE J211 filtering [19]. The measured within-session variance justifies the smaller count in later batches (Section 3.3). We characterize each of the two per-specimen objectives, the transmissibility after channel-frequency-class (CFC) 180 filtering and the mass-weighted rebound index, for repeatability before they enter the optimizer.

(3) An experiment-driven BO loop, built on BoTorch/Ax [20, 21] with a fully Bayesian sparse axis-aligned subspace surrogate (SAASBO) [22], that ingests the physical drop measurements, models per-print mass explicitly, and recommends each next batch of designs to fabricate. We report all four measured batches (35 specimens). The final Pareto front is entirely model-chosen, and the best-predicted design of the final batch had the lowest transmissibility measured in the campaign (Sections 3.4 and 4).

(4) A measured error budget for drop-tower ranking of printed specimens. We printed and tested one full nine-design batch two more times (360 additional drops). Print-to-print and seating variation exceed the within-session scatter, a broadband seat gauge (Section 3.3) flags the largest single-print errors, and we adopted a between-seat replication rule in response (Section 4.2).

(5) A simulation-only benchmark of multi-objective BO against uninformed sampling at a matched budget (Sections 3.5 and 4.4). On a MuJoCo drop-tower model, BO reached 93% of a best-known reference hypervolume after 45 designs, against 77 to 79% for four uninformed sampling baselines, and finished ahead on all ten seeds.

The completed evidence in this paper covers only single-cell, fixed-topology T_3-prism-inspired specimens under one *axial* drop condition. We plan quasi-static testing as an extension for future work. Off-axis and combined loading, controlled fatigue and damage-accumulation testing, multi-cell behavior, and landing-attitude variability are likewise future work (Section 5). We limit the novelty and generality claims above to this restricted prototype design space.

We organize the remainder of the paper as follows. Section 2 reviews tensegrity mechanics, multi-material 3D printing for energy absorption, and Bayesian optimization with an emphasis on multi-objective and noise-robust acquisition functions. Section 3 introduces the design parameterization, fabrication workflow, drop-tower protocol and objectives, BO loop, and the replicate rank-stability study. Section 4 reports all four measured batches, the surrogate audits, the replication findings, and the results of the simulation-only campaign of Section 3.5, and Section 5 discusses the implications and limitations of the approach. Section 6 concludes the paper.

2 Background

2.1 Tensegrity-Inspired Architectures for Energy Absorption. Tensegrity prisms and their planar extensions exhibit a softening-stiffening response under compression [3–5], which is desirable for impact mitigation because the structure can absorb a large fraction of the input energy at a controlled force plateau. Santos [23] characterized a tensegrity energy-dissipation metamaterial with equivalent viscous damping of ~23% and negligible residual strain per impact, which suggests that tensegrity-inspired architectures suit repeated-impact use. The space program literature documents large-scale tensegrity-inspired absorbers for planetary landing [6–10] and a long history of crushable energy-absorbing landing systems [24–26]. Pajunen et al. [11] directly 3D-printed truncated-tetrahedral tensegrity-inspired unit cells and measured load-limiting plateaus, which motivates the present focus on additively manufactured PLA/TPU composites. Davami et al. measured the dynamic energy absorption of additively manufactured, shape-recovering lattices [27], and Intrigila et al. fabricated and characterized a bistable tensegrity-like unit cell for lattice metamaterials [28], the closest published analog to the present cells. Neither study used multi-material FDM or closed-loop design optimization. Table S1 of the Supplementary Information positions the present work against selected prior efforts spanning the three elements combined here (impact-tested printed tensegrity-inspired structures, rigid-flexible fabrication, and BO of architected materials). None of the studies we reviewed combines a multi-material

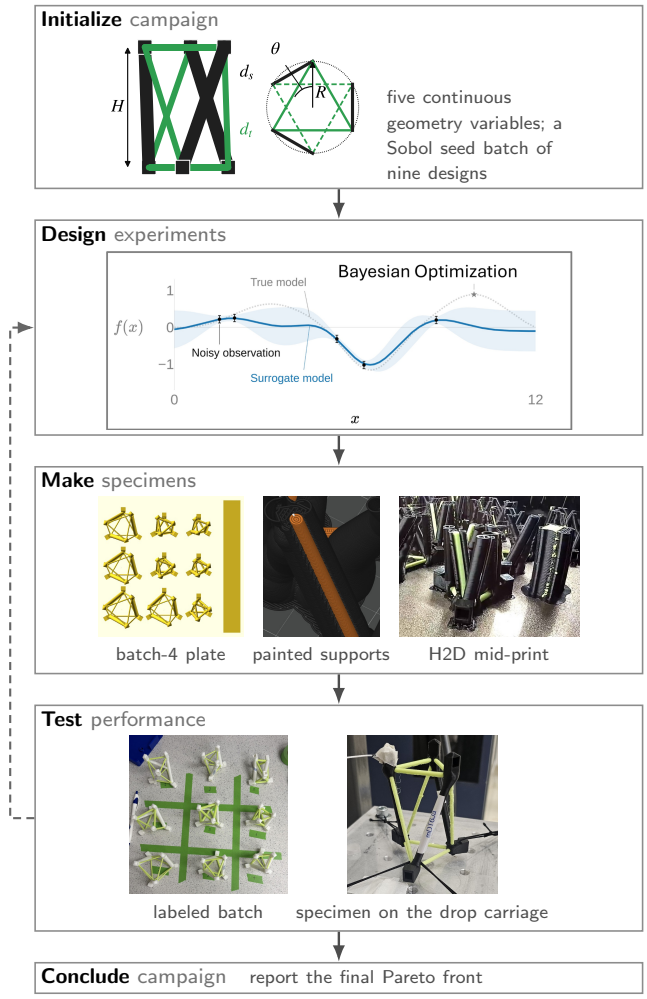


Fig. 1 Closed-loop, experiment-driven design framework. Initialize and Conclude run once, and the dashed arrow returns each tested batch to the Design step. *Initialize:* the five-variable T_3-prism design space of Table 1 (Fig. 2). *Design:* a Gaussian-process surrogate fitted to the noisy measurements recommends the next batch (Section 3.4). *Make:* we scale each batch to a common mass and lay it out as one build plate. We slice the plate with PLA supports painted by hand along the TPU tendons and print it in one dual-nozzle job, with the hard material (PLA) from the left nozzle and the soft material (TPU) from the right (Section 3.2). *Test:* we drop-test the labeled specimens one at a time on the drop-tower carriage (Section 3.3). *Conclude:* report the final Pareto front. The plate render is batch 4, the mid-print frame is a later replicate plate, and the labeled specimens are batch 2.

FDM tensegrity-inspired architecture with an *experiment*-driven optimization loop.

2.2 Multi-Material 3D Printing of PLA/TPU Composites. Multi-material rigid-soft FDM enables stiff/compliant combinations on a single platform with tunable energy absorption, demonstrated for thick-panel origami in which TPU hinges join the PLA panels [12] and for ABS/TPU honeycombs [13]. Neither study is itself a tensegrity, but both establish the rigid-soft co-printing route we adopt. Ye et al. [12] introduced a *wrapping-based* strategy in which continuous soft skins encapsulate rigid cores, reported to reduce interface delamination under cyclic loading, something we take inspiration from in our designs. Beyond these rigid-soft demonstrations, several studies show that additive manufacturing can produce tensegrity-inspired architectures with tunable mechanical response. Bauer et al. [29] showed that 3D-printed tensegrity metamaterials delocalize deformation to resist catastrophic failure, while Pajunen et al. [30] tuned acoustic bandgaps in 3D-printed tensegrity-inspired lattices through pre-strain, and Sabouni-Zawadzka et al. [31] experimentally characterized 3D-printed tensegrity-inspired metamaterials built on the simplex module. More directly relevant to impact, Almeida et al. [32] and Davami et al. [33] characterized the high-strain-rate and dynamic response of additively manufactured tensegrity-inspired structures, and Wang et al. [34] reported an integrated fabrication-and-validation workflow for tensegrity-inspired rigid-flexible metamaterials. None of these studies combines multi-material FDM with an experiment-driven optimization loop. Work on multi-material PLA/TPU sandwich and layered composites [35] reports quantitative bounds on stiffness, strength, and energy absorption for the same PLA/TPU pair used here, while FDM PLA fatigue data [36, 37] can inform conservative design bounds for the rigid struts. Yavas et al. have designed architected multi-material lattices with co-printed compliant skins for high specific energy absorption [38]. Recent surveys review additively manufactured polymeric energy absorbers [39, 40].

2.3 Bayesian Optimization for Architected-Material Design. Bayesian optimization is a sequential, model-based strategy for optimizing expensive black-box functions using a Gaussian-process surrogate [41, 42]. Modern implementations such as BoTorch [20] and the Ax platform built atop it [21] support parallel batched evaluations and many acquisition functions. Recent algorithmic advances include log expected improvement (log-EI) for numerical robustness [43], noisy expected hypervolume improvement (qNEHVI) for multi-objective settings [44], input-noise-robust acquisitions [45], and evolution-guided constrained multi-objective BO for self-driving labs [46]. Mixed/categorical search spaces, needed for the planned connectivity-topology extension rather than the continuous T_3-prism prototype studied here, have dedicated treatments in Bayesian optimization of combinatorial structures (BOCS) [47] and the one-hot-with-rounded-kernel construction of [48]. Prior work has applied BO to architected-material design [17] and to additive manufacturing [18]. Mo et al. [17] in particular established a multifidelity BO framework for architected materials by fusing low-fidelity simulations with high-fidelity ground truth via nonlinear information-fusion priors [49]. We shift the emphasis from simulation surrogates to direct experimental data, which suits materials whose constitutive response is difficult to calibrate from first principles.

3 Materials and Methods

3.1 Design Parameterization. We parameterize a family of tensegrity-inspired unit cells via four groups of design variables. For the T_3 prism that defines the initial BO search space (Section 3.4), only the first two groups are active and all of their variables are *continuous*. We hold the connectivity-topology and tiling groups fixed for this prototype and list them here as planned extensions of the design family:

- **PLA compression members.** Strut diameter d_s and length ℓ_s, with the slenderness ratio ℓ_s/d_s bounded away from buckling-dominated regimes.
- **TPU tension elements.** Cross-sectional area A_t (or equivalent diameter d_t for circular cross-sections).
- **Connectivity topology (future extension).** Selected from a discrete set of candidate unit cells (e.g., truncated-tetrahedral, prism-based, and octahedral variants). When activated, the design vector would encode this as a categorical choice. We hold it fixed at the T_3 prism for the present campaign.

- **Unit-cell tiling (future extension).** Number of cells $N_x \times N_y \times N_z$ and the orientation of each cell relative to the impact direction. We hold it fixed at a single cell here.

Design family and search-space evolution. We validated the fabrication and test pipeline on the first instance of this family, a three-strut, T_3-prism-inspired cell scaled to a ~50 mm bounding box. In batches 1 and 2 the BO search space has five continuous design variables over the ranges listed in Table 1. They are the base radius R, cell height H, twist angle θ, a single strut diameter d_s, and a single cable (tendon) diameter d_t. We treat the cable diameter d_t as a *continuous* variable on $[3.0, 5.5]$ mm. From batch 3 the space grows to include the print process: two per-specimen sparse-infill densities (strut and tendon, each 12 to 35%) that the acquisition chooses, and four filament settings (PLA and TPU nozzle temperature and maximum volumetric speed). The filament settings are physically one value per print job, so we draw them as one quasi-random point per batch, and they remain confounded with batch until a round varies them within a plate (Section 3.4). Under the prism's D_3 symmetry all struts are equivalent, and we also give all cables one diameter. We therefore use a single strut-diameter axis and a single cable-diameter axis, leaving room to relax to per-orbit (saddle, top, bottom) or fully per-member diameters once we have characterized the individual orbits. We hold the joint diameter d_j fixed at 7.0 mm. It sets the printed strut-end cage that houses the internal cable anchors (Section 3.2), so the TPU-tendon outlet count and the dual-nozzle clearance, rather than energy-absorption performance, primarily constrain it. Fixing it keeps the anchor geometry constant across specimens so that joint-strength changes do not confound the optimized variables. We defer relaxing d_j to a follow-on joint-design study. Figure 2 shows the design variables on a mid-range prism, six tested specimens drawn to one scale, and an as-printed specimen.

Constant-mass projection and printability screens. Because cell mass varies severalfold across the raw design box, an unconstrained search would partly reward designs for carrying more material rather than for their architecture. We therefore project every candidate onto a constant material budget before printing. The projection uniformly scales each design (all dimensions except the twist angle) until it meets its mass target, converged to within 0.15 g on rendered geometry. The target evolved with the campaign. Batches 1 and 2 held *solid* CAD mass $\rho_{\mathrm{PLA}} V_{\mathrm{PLA}} + \rho_{\mathrm{TPU}} V_{\mathrm{TPU}}$ at $m^\star = 30.95$ g, the solid mass of the reference prism. That budget does not hold printed grams constant, because PLA struts print at partial infill while thin TPU cables print near solid. As-printed masses spanned 18.50 to 22.04 g across the tested seed specimens, a coefficient of variation (CV, the standard deviation divided by the mean) of 5.5%, and 17.91 to 23.47 g (CV 8.7%) across batch 2. For those batches printed mass is therefore only approximately constant, and the surrogate carries each specimen's weighed mass explicitly (Section 3.4). From batch 3 the projection instead holds *printed* mass at 20.23 g, the mass of the printed reference specimen. A calibrated model predicts each design's printed mass from its geometry, and the projection scales the design until that prediction equals the target (Supplementary Information). The printed batches came in slightly light. Their mean masses were 0.2 to 0.6 g (1 to 3%) below the target, and the CV within a batch was 1.7% for batch 3 and 2.3% for batch 4.

We also checked each projected design against two printability limits, a 250 cm^3 size limit on $\pi R^2 H$ and a 3.0 mm minimum on the printed TPU cable diameter. We recorded both as warnings, and we used neither to reject or change a design. Because the projection scales the whole cell, nine specimens printed with cables of 2.51 to 2.78 mm, including all four batch-4 specimens that reduced the peak shock, and all nine printed. The screens and the defect record behind them are in the Supplementary Information (Section S2).

Table 1 Design variables for the T_3-prism campaign, as the BO code implements them. The seed batch is an $n = 9$ Sobol draw (fixed seed) over the five geometry variables, and we then project each design onto the constant-mass budget of the text. The process variables joined the space in batch 3. The acquisition chooses the two infill densities per specimen. The four filament settings admit one value per print job, and we therefore draw them as a single quasi-random point per batch, confounded with batch until a round varies them within a plate. We hold the joint diameter fixed. All specimens print upright on a Bambu Lab H2D, with the prism axis normal to the build plate and the base triangle on the bed (Fig. 1, Make panel). The seed batch used one specimen per named build plate, and each later batch used one nine-specimen plate.

Variable	Symbol	Range	Units	Active
Base radius	R	25-40	mm	batch 1 on
Cell height	H	60-110	mm	batch 1 on
Twist angle	θ	40-80	deg	batch 1 on
Strut (PLA) diameter	d_s	6.0-12.0	mm	batch 1 on
Cable (TPU) diameter	d_t	3.0-5.5	mm	batch 1 on
Strut infill density	-	12-35	%	batch 3 on
Tendon infill density	-	12-35	%	batch 3 on
PLA nozzle temperature	-	212-228	°C	batch 3 on[a]
PLA volumetric speed	-	20-30	mm^3/s	batch 3 on[a]
TPU nozzle temperature	-	235-245	°C	batch 3 on[a]
TPU volumetric speed	-	2.2-3.6	mm^3/s	batch 3 on[a]
Joint diameter (fixed)	d_j	7.0	mm	-

[a]One value per print job (a plate carries a single filament configuration), so per-batch rather than per-specimen.

As-built resolution of the continuous variables. Although we treat d_s and d_t as continuous, the nozzle diameter, the slicer line width, and support interaction discretize the as-built cross-sections, so the effective search granularity is coarser than the nominal continuous box. The realized member diameter changes in steps set by the slicer line width, and these steps determine the smallest CAD change that gives a distinguishable as-built member. We weigh every specimen on a 0.01 g scale, and mass is the only as-built quantity we measure for every specimen. We have not yet measured the dimensions of the printed members, so the geometry we attach to the surrogate is the commanded post-projection geometry. Repeated prints bound the fabrication scatter in mass. One seed design printed three times has a mass standard deviation of 0.457 g, the value we use in the observation noise model (Section 3.4). The three prints of the batch-3 plate differ by a per-print flow offset of up to 0.48 g rather than by random scatter (Section S2).

3.2 Multi-Material Fabrication.

We fabricated the specimens on a multi-material FDM printer using PLA for the rigid struts and TPU for the soft tension elements. The Make panel of Fig. 1 shows the steps from the build plate through slicing to the print. Rather than wrapping the struts in a continuous TPU skin, we anchored the TPU tension elements *inside* the ends of each PLA strut. The cables converging at a given strut end join within the strut, which acts as a rigid cage housing the junction, and the individual tendons then exit through discrete outlets. This keeps the soft-rigid interface in internal, compression-dominated pockets rather than at an exposed overmolded surface. This construction inverts the material roles of the wrapping strategy of Ye et al. [12], in which a soft skin covers a rigid core. Here a rigid PLA shell captures a soft TPU core, so that work is a precedent for multi-material co-printing but does not validate the present junction.

Print platform, joint, supports, and process control. We printed every specimen on a Bambu Lab H2D printer, which permits a single rigid-soft build without filament swaps. The strut-to-cable junction is a captive TPU core inside a PLA shell (Design F of the joint study in Section S3), which keeps the soft tendon away

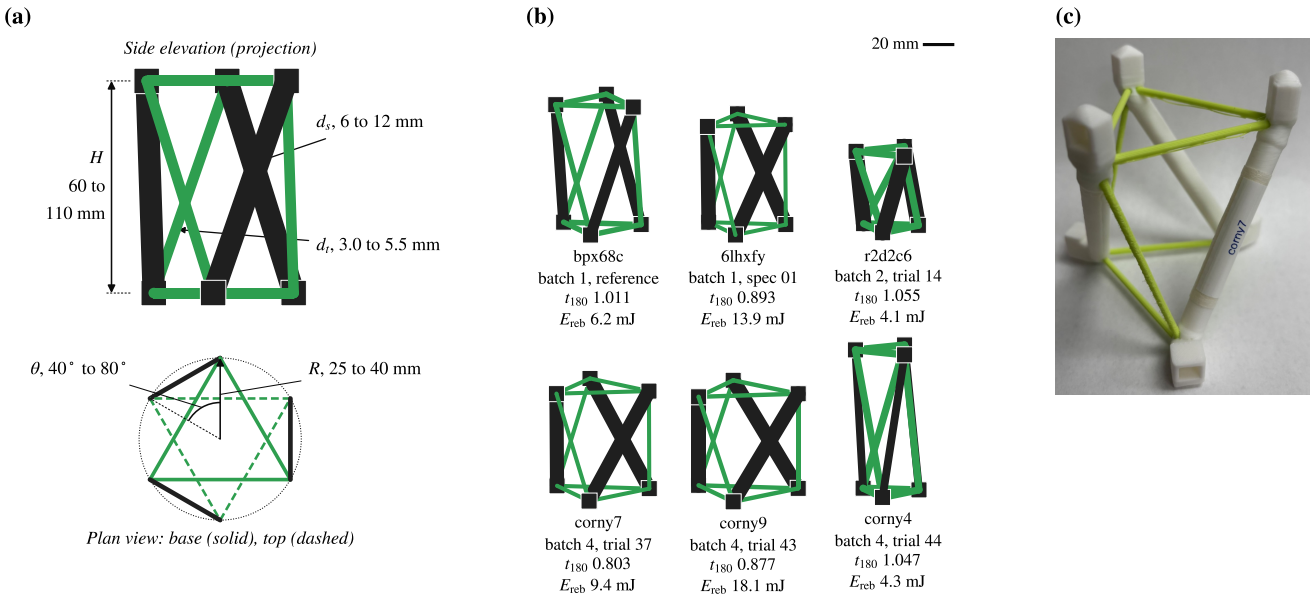


Fig. 2 Design space of the T_3-prism campaign. ***(a)*** **A mid-range design of the five-variable box (R = 32.5 mm, H = 85 mm, θ = 60°, d_s = 9 mm, d_t = 4.25 mm, before scaling to the mass target), labeled with the bounds of each variable. The side elevation is an orthographic projection, so each end triangle appears as a line, and H is the vertical distance between the planes through the joint centers. The plan view shows the base triangle (solid), the top triangle (dashed), the circumradius R, and the twist θ between them.** ***(b)*** **Six tested specimens at their printed, post-projection geometry and one millimeter scale, labeled with the batch, the Ax trial (or the Sobol spec of a seed specimen), and the measured t_{180} and E_{reb}.** ***(c)*** **Specimen corny7 as printed, with its ID label around one strut. The drawings show PLA struts in black, TPU tendons in green, and the square strut-end housings. We printed specimen corny7 in white PLA.**

from layer-line failure at the transition. Because near-vertical TPU tendons are otherwise unsupported during printing, we disabled automatic support generation and painted PLA edge supports manually along each tendon in the slicer. Support removal is the main source of the cosmetic defects we logged per specimen in the print key (Sections S4 and S6). TPU is hygroscopic, so we fed it from a heated dry box held near or below 8 % relative humidity, which we logged per print. Midway through the seed batch, a cold pull and a move to a dedicated high-flow 0.6 mm nozzle resolved a partial clog of the TPU extruder. The print key flags the affected specimens so that we can check any nozzle-related covariate against the measured objectives. Table S2 of the Supplementary Information lists the print parameters we transcribed from the archived as-printed slicer project (Section S5).

Figure 3 shows a full plate printing upright, with the painted supports rising beside the tendons.

3.3 Experimental Characterization.

We drop-tested every specimen on one instrumented fixture under one protocol. This section describes the fixture, the instrumentation and filtering, the two objectives computed from each session, and the replication checks that bound what a single session can show.

Drop-tower fixture. We characterized the specimens primarily by repeated drop-weight impact on a Lansmont Model 23 shock test system. The printed specimen rode upright on the free-fall carriage plate, which we raised 60 in (1.524 m) along guide rails and released. The plate landed on a single 0.5 in Shore 60 OO polyurethane cushioning sheet, which shaped the landing into a single sharp base pulse of roughly 1.6 ms duration. We selected this impact surface in a blinded crossover experiment over three candidate mat arrangements. Unlike a felt stack, it did not measurably compact over 100-drop sessions (we specified the seed sessions at 100 drops). We monitored rig health continuously through the impact velocity change Δv recovered from the base accelerometer. Over an uninterrupted 100-drop characterization session, the Δv coefficient of variation was 0.54% with no drift. We checked every campaign session against the healthy tower baseline. Details, including the within-session mat warm-up and recovery pattern, are in the Supplementary Information. One drop took roughly 42 s, so a seed-batch characterization took about 90 minutes per specimen and a later-batch session about 15.

Instrumentation and signal processing. A single-axis piezoelectric accelerometer on the carriage plate recorded the base input and triggered the acquisition. A triaxial accelerometer (Dytran 3133A4) seated in a printed pocket at the specimen's top vertex recorded the transmitted response. We cross-calibrated the two base-capable channels in place over 15 co-located drops (ratio 0.953 ± 0.001 at matched filtering) and archived the transducer settings with the data. We captured signals at 1.25 MHz over a 100 ms record with a 2 ms pre-trigger. We low-pass filtered all channels with the SAE J211/1 channel-frequency-class (CFC) filters [19], which we implemented from the standard's published coefficients, handling the two-pass corner frequency explicitly. A common implementation shortcut narrows the passband by roughly 20%, which we identified and corrected before the campaign. All reported numbers use the corrected filter. CFC-180 is the reporting class for peak metrics, and we retain CFC-1000 as a higher-bandwidth diagnostic. Figure 4 shows both channels of one drop at both filter classes.

Per-specimen objectives and session protocol. Session length changed as our understanding of the measurement noise improved, so the drop counts differ by batch. Seed sessions recorded 101 drops per specimen (103 for one specimen, autv5r). We discarded the first 2 drops of each session as seating warm-up, so 99 stabilized drops per seed specimen (101 for autv5r) enter each mean. A pre-campaign variance study on the commissioning datasets put the statistical minimum at five stabilized drops after two warm-ups. The seed sessions confirmed that the stabilized mean is insensitive to session length. Their within-session per-drop CVs were 0.17



(a) First layers (b) Mid-height (c) Final layers

Fig. 3 Three frames from the build-chamber camera timelapse of replicate plate 1 on the Bambu Lab H2D (https://youtu.be/EmCLms7-Bd8). The nine specimens print upright, with the prism axis normal to the build plate, beside a prime tower. (a) First layers, with the bottom TPU tendons and the strut-end housings. (b) Mid-height, with the struts and the painted PLA supports partly built. (c) Final layers, with the top tendons complete. We only crop and down-scale the frames.

to 0.48%, with flat post-warm-up trends. A replay of the seed sessions truncated to their first 20 stabilized drops left the transmissibility ranking of all nine tested seed specimens unchanged. The largest shift of a session mean was +0.008 (bag26v). Gaps between specimens adjacent in the ranking were 0.002 to 0.087, and at 20 drops the standard error of each mean (0.0002 to 0.0011) was still below the smallest gap. The closest pair (bpx68c and ajhby6) kept its order because truncation shifted both upward. We therefore shortened later sessions to about 20 drops per specimen. Batch-2 sessions recorded 21 or 22 drops per specimen and batches 3 and 4 recorded 20, so 18 to 20 stabilized drops per specimen remain after the same 2-drop warm-up discard. Exact per-session counts are in the archived per-drop records. Shortening the sessions cut tower time per specimen from about 90 minutes to about 15 while retaining roughly four times the minimum stabilized-drop budget. A shorter session enters the surrogate with a proportionally larger standard error, which the model downweights. The three prints of batch 3 (Section 4.2) later showed that seating and print-to-print variation limit what one session can claim far more than session length does, so we spent the freed tower time on re-seats. Each session yields two objectives, each reported as a stabilized mean with the per-drop scatter quoted as one standard deviation, and the standard error of the mean enters the surrogate's noise model (Section 3.4):

$$t_{180} = \frac{\max_t \|\mathbf{a}_{\text{top}}(t)\|_{\text{CFC180}}}{\max_t |a_{\text{base}}(t)|_{\text{CFC180}}}, \qquad (1)$$

$$e_{\text{reb}} = \frac{g\, t_{\text{second}}}{2\,\Delta v}, \qquad I_{\text{reb}} = m\, e_{\text{reb}}^2,$$

where t_{180} is the CFC-180 filtered peak-acceleration ratio, which we call the *transmissibility* throughout. It is the peak magnitude of the triaxial top-vertex response over the peak base input within the impact window. We use the term in the shock-testing sense that is standard for cushioning and packaging drop tests, a ratio of filtered peak accelerations across the specimen under a single transient, not the steady-state frequency-response function of vibration isolation. The two peaks need not be simultaneous, and the filter class is part of the definition. We cite SAE J211/1 only for the impact-channel filtering. Values above unity indicate the structure amplifies the peak shock reaching its top vertex; values below unity indicate attenuation.

Mass-weighted rebound index. The second objective is the *mass-weighted rebound index* I_{reb}, and lower is better, as for the transmissibility. It builds on the rebound fraction e_{reb}, the speed at which the top vertex hops after impact divided by the measured base velocity change Δv, that is, the fraction of the impact velocity returned at the top. We estimate the hop speed from the time to the second contact, t_{second}, assuming the hop is ballistic [50]. I_{reb} multiplies the square of e_{reb} by the specimen's weighed mass m, and we report it in milligrams. We adopted I_{reb} after the final batch, and any later batch minimizes it. The optimizer that recommended batches 2 to 4 used the first power of the rebound fraction instead, $E_{\text{reb}} = e_{\text{reb}}\, m\, g\, h$ with g the gravitational acceleration and h the drop height. The tables and figures report I_{reb}, except where they show the predictions the optimizer made at the time, which are in E_{reb}. Weighting by mass keeps grams in the objective, because equal rebound fractions on specimens of different mass are not equivalent outcomes for a payload. Where we quote rebound as a dimensionless descriptor, we use the complement of e_{reb}, the velocity loss index VLI $= 1 - e_{\text{reb}}$, the fraction of the impact velocity that the structure does not give back. Higher VLI is better, and across the 53 tested specimens it ranges from 0.937 to 0.986. The mass weighting matters most where printed mass varied. In batches 3 and 4, which held printed mass constant, ranking the specimens by I_{reb} and by e_{reb} gives nearly the same order (Spearman rank correlation $\rho = 0.998$ over 27 specimens, against 0.995 for E_{reb}). In batches 1 and 2 the two orders agree less well ($\rho = 0.917$ over 17, against 0.855).

Limits of the rebound measurement. Two caveats apply. First, the rebound fraction is a timing index. The specimen and the carriage fall together and share one arrival speed, but the carriage itself bounces back off the mat. High-speed video (959 fps) of seven batch-4 drops shows that bounce at 0.56 to 0.64 m/s, so Δv (5.1 to 5.4 m/s) is the arrival speed plus the bounce. This inflates Δv by about the same factor for every design and leaves their order unchanged. The same check showed that the second burst on the top channel is not always a hop. Replicate prints of two batch-4 designs (corny7 and corny8) never left the plate, and their second burst was the tendons pulling taut again. The rebound fraction should therefore not be read as a coefficient of restitution. Second, I_{reb} is a ranking score that penalizes designs, and grams, that send more motion back to the payload. It is not a measured energy (see the next paragraph). Once we validate the hop timing, the rebound fraction could become a constraint rather than an objective (Section 5).

Square of the rebound fraction. Returned kinetic energy scales with the *square* of a velocity ratio, so I_{reb} weights rebound the way an energy-based score would. It is still not an energy. Multiplied by gh, it would give the kinetic energy of the whole specimen at the hop speed, but only the top-vertex subassembly hops, and we have not yet validated the ballistic reading of the second contact. Squaring also halves the leverage of mass, because ranking by $m\, e_{\text{reb}}^2$ is the same as ranking by $\sqrt{m}\, e_{\text{reb}}$. The change of form moves the record only at the margins. Because gh is the same for every drop, E_{reb} and $m\, e_{\text{reb}}$ rank all 35 tested designs identically. Of the 595 pairs of designs, I_{reb} reverses the order of 21, and 16 of those 21 pairs differ by less than 5% in E_{reb} (Spearman $\rho = 0.990$ between the two design orderings). The final front of first prints is the same under both forms, and the design-mean front loses

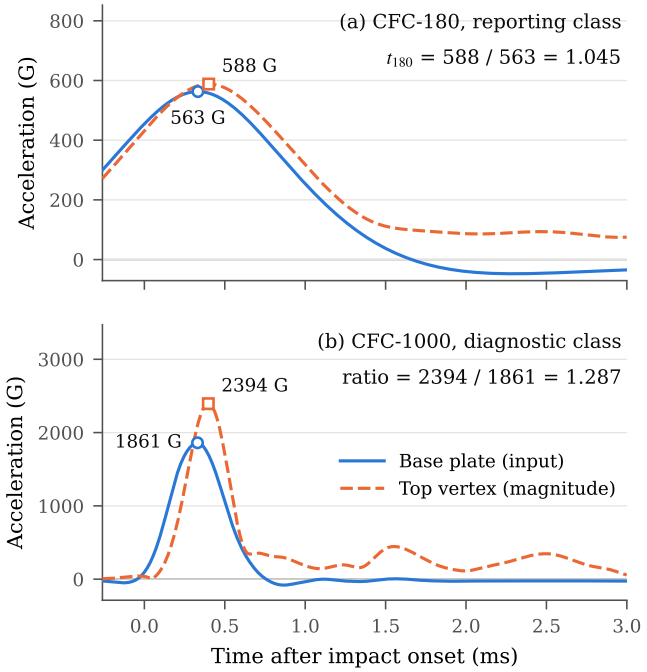


Fig. 4 Base-plate input and top-vertex response for one commissioning drop, shown to illustrate the transmissibility definition. The specimen is 57vqhx from the pre-campaign print-defect study (2026-07-28, event 50 of its session), dropped from 60 in onto the earlier stack of four felt pads and one cardboard sheet. Campaign drops land on the polyurethane surface instead. We low-pass filter both channels with the SAE J211/1 CFC filter, and the top-vertex trace is the magnitude of the three filtered axes. The zero-phase CFC-180 filter spreads the sub-millisecond pulse in both directions in time, so its traces are already nonzero at impact onset. (a) At CFC-180, the reporting class, the peak ratio gives t_{180} = 588/563 = 1.045. (b) At CFC-1000, the diagnostic class, the same drop gives 2394/1861 = 1.287.

r2d2c2 under I_{reb}. Eight of the 21 reversals come from averaging repeated prints of the batch-3 designs. The mean of the squares exceeds the square of the mean, so I_{reb} penalizes a design whose prints rebound inconsistently, a penalty that tracks print scatter as well as design performance. We archive the comparison and its per-design and per-specimen tables with the manuscript data (`scripts/analysis/pareto_rebound_index_variants.py`), and `scripts/analysis/rebound_index_numbers.py` recomputes every I_{reb} number quoted here after first reproducing the E_{reb} ones.

Session diagnostics and replication. The within-session per-drop CV of t_{180} spans 0.06 to 2.0% across all 53 tested specimens (0.17 to 0.48% in the seed batch). These values describe repeated drops of the same physical specimen in one seating and do not estimate print-to-print or seating-to-seating repeatability. Serial dependence among drops may also make $s/\sqrt{n_{\text{drops}}}$ optimistic. As an advisory check on how well a specimen is seated on the carriage, we computed a *seat gauge* for every session, the ratio of its CFC-1000 to its CFC-180 transmissibility. The CFC-180 filter keeps the slower part of the response (its corner is near 300 Hz). CFC-1000 (corner near 1.65 kHz) also keeps short, sharp spikes, such as a loosely seated specimen or top sensor knocking against its seat. A well-seated session gives nearly the same ratio at both filter classes, and the gauge read 1.00 to 1.07 on most sessions. We call a session that reads 1.15 or more a *re-seat candidate*: we should take its specimen off the carriage, mount it again, and re-test it before using its value for a decision. Two replication measurements bound what a single session can claim. The seed attenuator, re-measured across a day and an independent re-seat, agreed to 0.13%. Three prints of the batch-3 designs put the standard deviation between specimens and seatings at 0.0087 in t_{180}, once we set aside sessions that fail the seat gauge or carry a drift flag (Section 4.2). The two largest deviations, 10 and 20% above their design medians, both came with a failed gauge. We therefore adopted a between-seat replication rule: a single-seating value that would drive a decision (a batch extreme, a claimed attenuator, a front membership) requires confirmation on an independent re-seat before we treat it as a design property.

Planned quasi-static extension and complementary data streams. The drop test loads the specimens in their elastic regime. The crush lasts 1 to 3 ms and the specimen recovers, so no reprocessing of these captures can recover crush-energy metrics such as specific energy absorption (SEA) and compaction efficiency. We plan a quasi-static compression campaign on a benchtop load frame to supply them as a complementary characterization of selected designs, not as objectives of the drop-tower campaign. The metric definitions, the protocol we will follow, and its scope limits are in the Supplementary Information (Section S8). High-speed video (959 fps, three clips per specimen) documented the post-impact rebound and recovery, but it cannot resolve the millisecond-scale impact pulse, so the accelerometer channels alone produce the objectives.

3.4 Experiment-Driven Bayesian Optimization Loop. The optimization loop maintains a Gaussian-process (GP) surrogate over the design space defined in Section 3.1. After each round of physical experiments, we refit the surrogate and maximize an acquisition function to recommend the next batch of designs to fabricate. We pose the campaign as a two-objective minimization of the transmissibility t_{180} and the mass-weighted rebound index I_{reb} of Eq. (1). The optimizer recommended batches 2 to 4 under the first-power form E_{reb} in its place (Section 3.3). Minimizing either form favors a small rebound fraction on a light specimen. Both objectives serve the same payload. The transmissibility t_{180} bounds the peak shock the payload feels during the impact, and I_{reb} penalizes the motion thrown back into it afterward. The seed data showed an apparent trade-off, driven substantially by the single strongest attenuator and not statistically resolved at seven mapped specimens (Spearman $\rho = -0.39$, $p = 0.38$; Section 5). After four batches, the measured front is nondegenerate (its members span both axes), and the rebound fraction remains a candidate to become a constraint once we validate its ballistic reading. Any later batch minimizes I_{reb}. Every measured specimen has an I_{reb} value as well, so later results stay comparable with the existing front. We implement the loop in the Ax platform [21] over BoTorch [20], with noisy expected hypervolume improvement (qNEHVI) [44] as the multi-objective acquisition. The T_3-prism search space is fully continuous, including the process variables added in batch 3 (Table 1), so we need no categorical encoding.

Surrogate and noise model. The surrogate is the fully Bayesian sparse axis-aligned subspace model (SAASBO) [22]. It uses a Matérn-5/2 kernel with automatic relevance determination whose length scales carry strong sparsity-inducing priors, and it marginalizes those length scales by Markov-chain Monte Carlo with the No-U-Turn Sampler (NUTS) rather than point-estimating them. Every fit that generated a batch, and every archived prediction, used the BoTorch default budget of 512 warmup steps and 256 samples, thinned to 16 retained draws. The leave-one-design-out cross-validation of Section 4.3 refits each fold at that same budget, and the per-specimen leave-one-out refits used 256 warmup steps and 128 samples. Every refit in the campaign, through the batch-4 ingestion, uses this model with qNEHVI on top (Ax 0.4

for the early rounds, Ax 0.5.0 from the batch-3 refit). We retain SAASBO, even though the dimensionality is modest, because marginalizing the hyperparameters tends to improve uncertainty quantification at the very small sample sizes of physical campaigns. We audit the fitted model each round with cross-validation (leave-one-out, and leave-one-design-out once replicates entered the record) and length-scale-based sensitivity diagnostics (Section 4). We normalize inputs to the unit cube and standardize outputs. We supply observation noise per observation as a fixed variance. For t_{180} it is the squared standard error of the stabilized mean, $\sigma_t^2 = s_{\text{drop}}^2/n_{\text{drops}}$. For $I_{\text{reb}} = m\, e_{\text{reb}}^2$ it adds two relative errors in quadrature, $(\sigma_I/I_{\text{reb}})^2 = (2\sigma_e/e_{\text{reb}})^2 + (\sigma_m/m)^2$, the first-order propagation of its two inputs. The first term is the standard error σ_e of the session's mean rebound fraction, doubled because e_{reb} enters squared. The second term stands in for print-to-print variation at a fixed design. Here $\sigma_m = 0.457$ g is the standard deviation of printed mass over three prints of one seed design (spec 08, 21.42 to 22.29 g), about 2.3% of a typical specimen. A re-print that weighs more or less changes I_{reb} in proportion. The mass term is the larger of the two for most specimens. The refits through batch 4 used the same construction for E_{reb}, without the factor of 2. The estimand this noise model implies is the response of the tested specimen in its seating, not of a freshly printed specimen at the same design. The pre-registered review of the objective choices (Section 5) argued that the decision-relevant target is the latter. To account for that target, we use the measured replication evidence of Section 4.2 (second and third prints of one batch) together with the between-seat replication rule, rather than an inflated per-specimen noise term. We ingested the nine second-print specimens, re-prints of the same designs, as distinct completed trials at identical design coordinates, each with its own weighed mass and per-session standard error and with no pre-pooling. Later refits therefore observe measured design-level replicates directly. Because we supply per-session variances as known, no separately inferred noise parameter absorbs replicate disagreement. Instead, it widens the posterior through the fit itself. We tested the third print after the last refit reported here, and it has not yet entered the fit. SAASBO is the prespecified surrogate. We make no claim that it outperforms a standard single-task GP at this dimensionality and sample size, and the planned head-to-head calibration comparison against a point-estimated GP with the same kernel remains future work.

Mass treatment. Because the constant *solid*-mass projection of batches 1 and 2 does not hold printed grams constant (Section 3.1), printed mass correlates with the shape variables in the seed batch, and with t_{180} ($r = 0.83$, $n = 7$, dominated by the lightest specimen). We therefore keep mass explicit in the loop. We compute the second objective, I_{reb}, with each specimen's weighed mass, as we did for E_{reb}, so it penalizes added grams through the motion they send back to the payload. The transmissibility t_{180} stays a ratio on purpose, because payload peak acceleration is a damage criterion that does not scale with specimen mass. The objective surrogates take the design coordinates of Table 1 as their inputs. We attach printed mass as a separately modeled *tracking metric*, so the model learns as-printed mass as a function of the design coordinates and every recommendation reports a predicted printed mass. In the diagnostic refit behind the seed-state length-scale diagnostics (Supplementary Information), we additionally include printed mass as an input to probe this confound. With the calibrated printed-mass model (Supplementary Information), we then held *printed* grams constant in batches 3 and 4. Across all 53 tested specimens, printed mass explains about 1% of the variance in transmissibility ($r = 0.12$), so the experimental design, rather than an adjustment in the analysis, removed the seed batch's mass confound (Section 4.3).

Budget and seeding. We drew the nine seed designs as the first points of a scrambled Sobol sequence [51, 52], the initialization step of the Ax platform, with a recorded seed (0). A Sobol sequence is a quasi-random sequence that covers the five-variable box more evenly than the same number of independent random draws. We printed the nine designs together with the fixed reference prism (S0). Optimization proceeds in sequential batches of $q = 9$ prints, sized to one build-plate generation cycle. We report three model-recommended batches here (Ax trials 10 to 18, 28 to 36, and 37 to 45). The numbering gap records a superseded draft batch that we regenerated, before printing, when we simplified the batch-level filament handling. From batch 3 the four filament settings take one quasi-random point per batch, formally the settings of that batch's best-predicted specimen, so those axes are confounded with batch, and we report them as unidentified rather than learned. We record all random seeds, the full experiment state, and the recommendation tables per round. The campaign code carries full computational reproducibility (the exact processing pipeline and model settings). We plan to escalate to trust-region BO (TuRBO) [53] only if per-member-diameter extensions push the design space well beyond the present regime. We retain a single-objective log-Expected-Improvement run [43] on t_{180} as a baseline to quantify the benefit of the multi-objective formulation. The campaign code, the recommended batches, and the data behind every table and figure are in the project repository, which we will archive under a Zenodo DOI at acceptance (Data Availability Statement).

3.5 Simulation-Only Campaign. To test the optimizer on a problem whose front we can map, we also ran a campaign entirely in simulation. The simulator is a MuJoCo [54] model of the drop tower with rigid PLA struts, tension-only TPU tendons, and an impact-surface contact calibrated once against the reference prism (S0). It reports t_{180} with the bench's CFC-180 filter. We fix printed mass at 20.23 g, and the search varies four shape ratios, with printed tendon diameter of at least 3.0 mm and envelope volume $\pi R^2 H$ of at most 250 cm^3 as constraints. The two minimized objectives are t_{180} and the envelope volume, the bulk a lander must stow. Each of ten seeds spends 45 simulated designs, a nine-design Sobol batch and then four batches of nine chosen by qNEHVI on a single-task GP in Ax. Random, Sobol, Latin hypercube [55], and compass-search [56] baselines run at the same budget and seeds and skip unprintable candidates without simulating them. Section S14 of the Supplementary Information gives the simulator and scoring details.

3.6 Replicate Rank-Stability Study. Repeated prints of the batch-3 designs (Section 4.2) showed that a single specimen's measurement can move between seatings and prints by more than the design signal separating neighboring designs. To test whether the measured ordering survives fabrication and seating variation, we re-printed three batch-4 designs chosen before we printed any replicate. Trial 37 (corny7) was the batch-4 best-predicted design and measured $t_{180} = 0.803$, the lowest value of the campaign. Trial 39 (corny8) is its near twin in design space, differing mainly in TPU infill, yet it measured 0.150 higher. Trial 38 (corny2) comes from the tall, low-twist family at the far end of the batch-4 ranking. Each of three plates carries three copies of each design, printed from one G-code file at the batch-4 filament settings, for 27 specimens in total (Fig. 5).

We drop-tested plate 1 on 2026-10-17 under the standard protocol of Section 3.3, and the session recorded two instrument faults. For about an hour an intermittent fault on the base-input channel, which also triggers the acquisition, set off the trigger away from the impact, so it mistimed 56 captures, which hold no impact. The top-vertex accelerometer also knocked at impact onset in every session on the corny7 and corny8 copies, a spike that reads the transmissibility high and fails the seat gauge. On 2026-10-19 the input-channel instrumentation failed and drop testing stopped. We printed and weighed plates 2 and 3 but have not tested them. We withhold the replicate record until the repeats are complete on re-

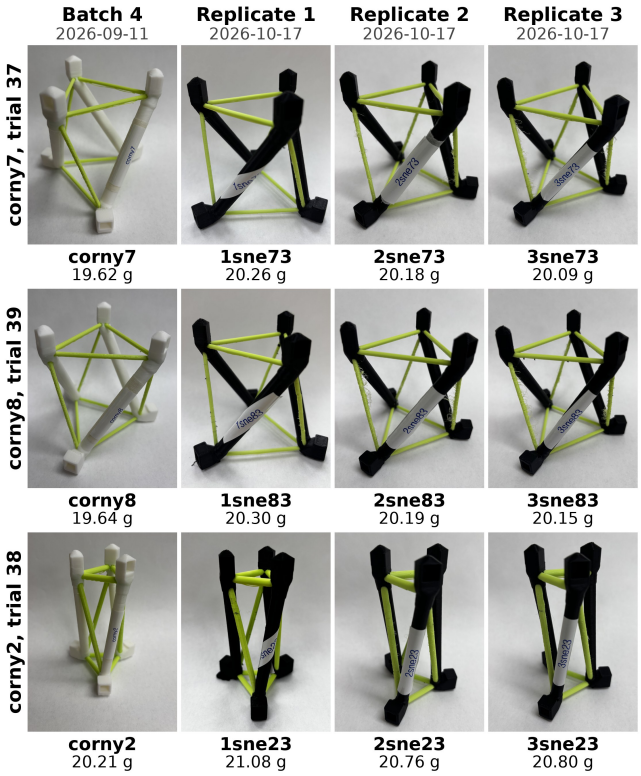


Fig. 5 Specimens of the three replicate-study designs, as photographed for the print log. Rows are trials 37 (corny7), 39 (corny8), and 38 (corny2). The first column is the batch-4 specimen (print log of 2026-09-11), and the other three are the front-row copy from replicate plates 1 to 3 (print log of 2026-10-17). Labels give the specimen ID and weighed mass. The batch-4 struts are white PLA and the replicate struts black PLA. Camera distance varies between photographs, so apparent size is not comparable across cells. The replicate drop tests are not complete, so the figure reports no measured values.

paired instrumentation, so no replicate value appears in this paper. The Supplementary Information gives the plate layout, the weighed masses, and the fault record (Section S10). We also plan a metal-analog comparison in which we rebuild worst-, mid-, and best-predicted designs from hollow aluminum rods and stainless-steel threaded cables and rank them against their PLA/TPU equivalents (Section S11).

3.7 Use of AI Tools. We used AI tools under our direction throughout this study and reviewed and edited their output at every step. The Acknowledgment names each tool and model and what we used them for. The repository file `manuscript/AI-USE.md` is the full inventory, including every automated session run on this manuscript (Data Availability Statement).

4 Results

Figure 6 shows every specimen measured in the four batches and the non-dominated front after each batch. Section 4.1 reports the seed batch and Section 4.2 the three model-recommended batches.

4.1 Seed-Batch Measurements. We printed all nine Sobol specimens and the reference prism and declared them acceptable to test. Nine specimens (seven mapped Sobol designs, the reference prism, and one specimen whose design mapping is unconfirmed) completed seed characterization sessions, totaling over 900 clean captures within a project-wide history of more than 4,000 instrumented drops. We printed the two remaining seed designs (specs

Table 2 Seed-batch drop-tower results: 99 stabilized drops per specimen (101 for autv5r) at 60 in onto the standardized polyurethane surface, CFC-180 filtering, counted after we discarded the first 2 warm-up drops of each session. Means ± one standard deviation over the stabilized drops. We order the rows by t_{180}. I_{reb} uses each specimen's weighed mass per Eq. (1). A dash denotes a quantity that we cannot compute from the available record. We exclude specimen *amdjwm*, which lacks a weighed mass and a confirmed design mapping, from the surrogate fit and from all mass-dependent quantities. The ringdown frequencies of these sessions are in the Supplementary Information. We printed specs 03 and 06, but they have no drop sessions in the campaign record. We printed all specimens upright (Fig. 3).

Design	Specimen	Mass (g)	t_{180}	I_{reb} (mg)
Spec 01	6lhxfy	18.50	0.8931 ± 0.0042	46.9
(unmapped)	amdjwm	-	0.9805 ± 0.0025	-
Spec 05	6nheas	21.73	0.9970 ± 0.0032	35.2
S0 ref.	bpx68c	20.23	1.0111 ± 0.0024	8.5
Spec 07	ajhby6	20.53	1.0127 ± 0.0038	8.1
Spec 00	9hhbkp	21.62	1.0183 ± 0.0017	10.0
Spec 04	nvxsrv	20.66	1.0275 ± 0.0044	14.6
Spec 02	autv5r	22.04	1.0404 ± 0.0035	15.8
Spec 08	bag26v	21.42	1.0616 ± 0.0051	12.4

03 and 06), but their drop sessions are absent from the campaign record and they do not enter any fit or figure. Table 2 reports the stabilized objectives. Three findings frame the batch. First, the measured t_{180} spans 0.893 to 1.062. Six of nine tested specimens *amplify* the peak shock reaching the top vertex (means above unity), and two sit marginally below unity (0.997 and 0.980). One tested specimen (the print of Sobol spec 01, the thick-strut, thin-cable, high-twist corner of the batch) had a mean t_{180} of 0.893, indicating peak attenuation under the tested condition. This specimen later reproduced its attenuation within 0.13% across an independent re-seat. It is the campaign's positive control for the replication rule of Section 3.3. Across the tested specimens, t_{180} spans 0.169 (18.9% of the observed minimum). This between-specimen range far exceeds the 0.17 to 0.48% within-session per-drop CV, but one mechanically tested specimen per geometry cannot separate geometry effects from print-to-print variation or other fabrication covariates. Second, the objectives show an apparent trade-off in this sample. The best attenuator has the lowest velocity loss index of the batch (0.950) and the largest I_{reb} (46.9 mg), so the observed front is nondegenerate. The trade-off rests largely on that single specimen, though, and the data do not statistically resolve it at this sample size (Section 5). Third, among the eight mapped specimens with weighed masses, printed mass and t_{180} showed a large sample correlation ($r = 0.82$, $n = 8$). The association is confounded with the design coordinates and is highly sensitive to the lightest specimen ($r = 0.26$ when we omit 6lhxfy), so it is not evidence of a causal mass effect. It nonetheless motivated the mass-aware objective formulation of Section 3.4, and the constant-printed-mass batches later removed the confound (Section 4.3). Figure 7 renders the seed geometries to a common scale, showing how widely the constant-mass box spans aspect ratio, member sizing, and twist.

4.2 Model-Recommended Batches and Replication. Table 3 reports the stabilized objectives for the three model-recommended batches, and Fig. 6 overlays every measured specimen with the non-dominated front after each batch. We evaluate Pareto membership over specimens with both objectives and omit the unmapped seed specimen amdjwm, whose I_{reb} we cannot compute without a weighed mass. Dominated hypervolume, evaluated at the fixed reference point $(t_{180}, I_{reb}) = (1.35, 50\,\text{mg})$, grew from 14.72 after the seed batch to 16.98 after batch 2 (+15.3%), 17.13 after batch 3 (+0.9%), and 23.02 after batch 4 (+34.4%). That is a 56% total gain over the seed front. The reference sits just

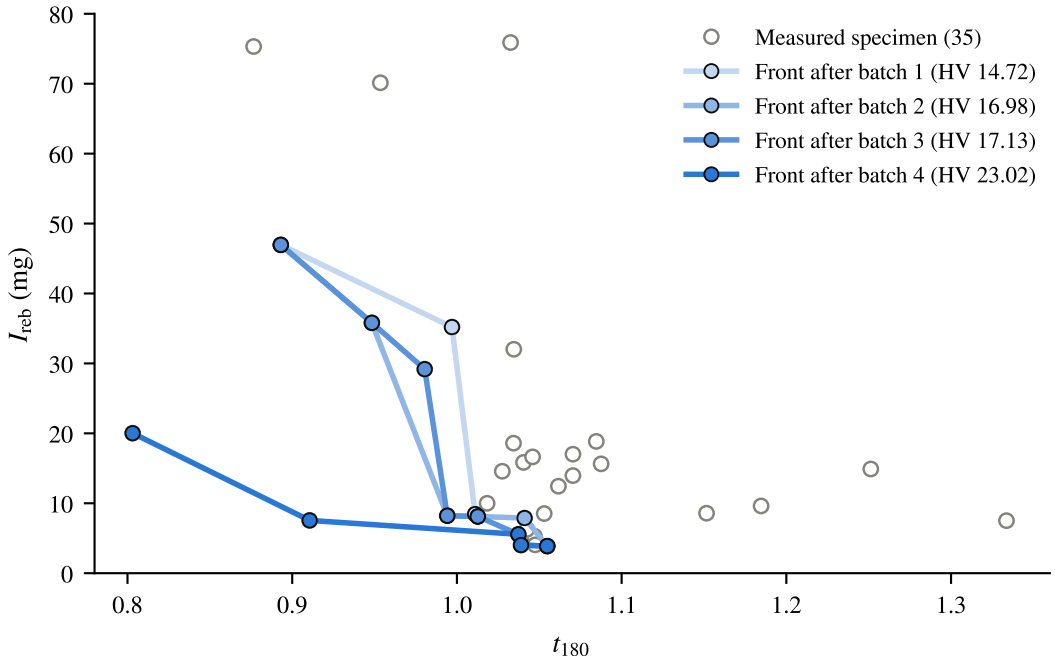


Fig. 6 Measured objective space and front evolution across the four batches. The vertical axis is the mass-weighted rebound index $I_{reb} = m\,e^2_{reb}$ of Eq. (1). Gray circles are all 35 measured specimens at their stabilized means. The blue lines are the non-dominated fronts after each batch, with the dominated hypervolume (HV) at the (1.35, 50 mg) reference in the legend. Batch 2 grew the front on the I_{reb} axis, batch 3 filled the knee, and batch 4 extended it to lower transmissibility. We draw repeated batch-3 designs at their first-print values. The text gives the design-mean front. Lower is better on both axes.

beyond the worst specimen on each axis through batch 2. The (1.35, 15 mJ) reference for E_{reb}, the optimizer's own form, follows the same rule, and under it the gain is 65%. We have not run a budget-matched physical baseline of Sobol designs, so the comparison the data support is against the campaign's own Sobol seed. After batch 4, every member of the combined front (corny7, corny6, drran5, drran3, r2d2c6) is a model-recommended design, and the last two seed specimens (6lhxfy and ajhby6) left the front when the batch-4 record dominated them. These fronts score every specimen as measured, with each repeated batch-3 design at its first print. Scoring each batch-3 design instead at the mean of its screened prints (Table S6 of the Supplementary Information) removes drran5 and drran3 from the final front. The design-mean front is corny7, corny6, corny4, and r2d2c6, still entirely model-recommended. Under E_{reb} it also holds r2d2c2, whose membership is provisional on its drift flag. Scored the same way, batch 3 added 0.1% to the hypervolume instead of 0.9%, so most of its apparent gain came from favorable first prints.

Batch 2 (Ax trials 10 to 18). We generated the first recommended batch from the seed fit and printed it under the constant-solid-mass projection, so its printed masses still spread widely (17.91 to 23.47 g). We archive a regenerated constant-printed-mass recommendation set alongside the as-printed one, and every result reported here is for the as-printed set. The batch grew the front almost entirely on the I_{reb} axis, and four batch-2 specimens joined the front. Specimen r2d2c6 still holds the lowest I_{reb} on record, 3.9 mg, and its velocity loss index of 0.986 ties two later specimens (drran3 and corny4) for the highest. On the transmissibility axis, the archived posterior predictions sat below the measurements for nine of nine specimens, with a mean absolute percentage error (MAPE) of 14.4%. The worst case was r2d2c3 (predicted 0.910, measured 1.334), the strongest amplifier in the campaign record. That nine-of-nine optimism probably widened the refit posterior for every batch that followed.

Batch 3 (Ax trials 28 to 36). Batch 3 was the first batch at constant *printed* mass (19.24 to 20.06 g, CV 1.7%) and the first with the infill and filament axes active (Section 3.1). Its first prints filled the knee of the front without extending it (+0.9% hypervolume, three specimens joined). Eight of nine specimens measured within 0.3 posterior standard deviations of their predictions (MAPE 3.8%). Two first-print results stood out. The best-predicted design (trial 32, predicted 1.020) measured 1.251, then the strongest amplification on record, and the batch's only attenuator (drran8, 0.980, drift-flagged) was its only twisted design. Neither result held up on the later prints described below.

Batch 4 (Ax trials 37 to 45). The acquisition split the batch between the wide, high-twist, thin-cable family it had been steering toward since batch 2 and a previously unvisited tall, low-twist corner. Four of nine specimens attenuate, all of them from the wide family. Specimen corny7, the batch's best-predicted design (trial 37, predicted 0.934 ± 0.13), measured t_{180} = 0.8031 ± 0.0025, the strongest attenuation in the program, 10% below the best seed specimen and 21% below the reference prism. Its velocity loss index, 0.968 (I_{reb} = 20.0 mg), is mid-pack, and it dominates the seed attenuator on both objectives (Fig. 8). Within the attenuating family, the archived predictions ranked the four measurements perfectly (Spearman ρ = +1.0). Across all nine specimens the rank correlation is 0.82, although the five amplifier predictions span only 0.9% and so carry no ordering signal. The batch MAPE was 6.0%, and no specimen left its predictive band by more than one standard deviation. The record itself beat its band by 1.0 standard deviation in the good direction. Per the between-seat replication rule, the batch-4 attenuation values, including the record, are single-seating measurements whose re-seat confirmations we have scheduled. One session (corny5) carries a drift flag, so its mean is provisional. Another (corny6) was erratic, with a per-drop CV of 2.0%, but attenuates under any reading of its record.

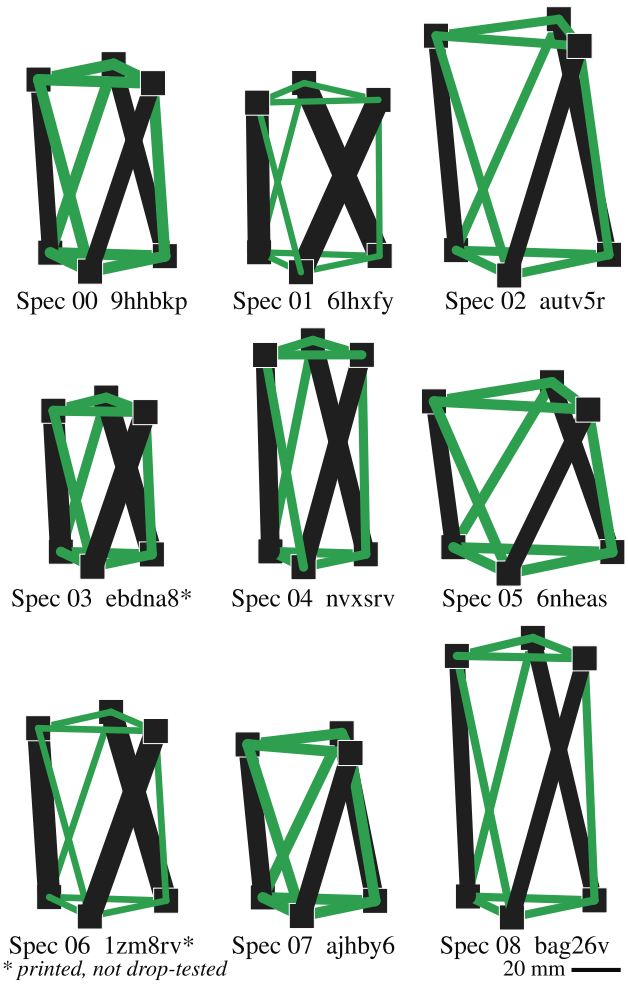


Fig. 7 The nine Sobol seed designs at their printed geometry, drawn to one millimeter scale as in Fig. 2(b). We label each drawing with its Sobol spec and the ID of its printed specimen. We printed specs 03 and 06 (asterisks) but did not drop-test them, and we label spec 08 with its tested specimen, bag26v.

Replication: three prints of batch 3. We printed the batch-3 plate three times and drop-tested every specimen under the identical protocol, 540 drops in all. In the later prints, specimens 2dran*N* and dran3*N* re-print drran*N* at the same Ax trial (round-3 print key). Each of the nine designs therefore has three specimens (Fig. 9; we tabulate the per-print values in Table S6 of the Supplementary Information). The three prints averaged 19.59, 19.86, and 20.07 g. Neither extreme of the first print recurred. Trial 32, whose first print (drran7) measured 1.2513, then the strongest amplification on record, measured 1.0286 and 1.0448 on its later prints, close to its prediction of 1.020. The drift-flagged 0.9804 attenuator of trial 35 (drran8) measured 1.0109 and 0.9958. Trial 35 is the lowest-transmissibility design in every print, but its screened mean is 1.003, so batch 3 contains no reproducibly attenuating design.

We screen each session with the seat gauge of Section 3.3, and every large same-design deviation came with a failed gauge. The four sessions that read 1.15 or more were each their design's highest value, 4.5 to 19.8% above the design's median over the three prints. The 23 sessions below that threshold sat a median 0.2% from their design medians (at most 3.7%, on a drift-flagged session), and the gauge correlates with the deviation (Spearman $\rho = 0.52$, $p = 0.006$). The third print supplied one new gross outlier, dran35 (trial 28), at 1.168 with a gauge of 2.43, against 1.037 and 1.060 for its earlier prints. Its re-seat is pending, and the screened means leave it out.

Setting aside the four gauge failures and the two drift-flagged sessions leaves 21 of the 27. Fitting those 21 with one mean per design plus one offset per print leaves a residual standard deviation of 0.0087 in t_{180} (10 degrees of freedom). That is below the 0.013 estimated from the first two prints alone, and still an order of magnitude above the within-session standard errors (median 0.0006) that the surrogate ingests. The second print's offset is +0.008 and the third's −0.003, so the upward shift of the second print did not recur. The third print was also the heaviest, so that shift was probably not a mass effect. Over the screened sessions the intraclass correlation of t_{180} (the share of its variance that lies between designs) is 0.58, against 0.06 when we include every session. Most of the design signal in batch 3 comes from trial 35. Without it, the other eight design means span 1.028 to 1.045 (standard deviation 0.006), less than the scatter of a single specimen. Single specimens therefore cannot rank those eight, and the orderings from different prints agree poorly (Spearman ρ of 0.08, 0.67, and 0.20 between pairs of prints). The frozen batch-3 predictions rank the nine design means with $\rho = 0.75$ ($p = 0.02$) when we adjust each mean for the print offsets, and with $\rho = 0.55$ ($p = 0.13$) without that adjustment. Against single prints they reach 0.20, 0.72, and 0.30.

I_{reb} repeats far less well than transmissibility. For each design the largest of its three values is 1.9 to 8.3 times the smallest (6.3 against 52.1 mg at trial 30). The square compounds the scatter of e_{reb}, and the same ratios are 1.35 to 2.9 under E_{reb}. The rank correlations between prints are −0.18, 0.25, and 0.33 under either form, and the scatter between prints (standard deviation 17.0 mg) exceeds the spread of the design means (10.2 mg).

The 1.2513 session had passed every within-session stability check (per-drop CV 0.77%, no drift), so a stable session is not enough to show that its value reflects the design. We adopted the between-seat replication rule of Section 3.3 in response. Across an independent re-seat, the seed attenuator agreed to within 0.13%, so genuine attenuation can reproduce. The seat gauge accompanies the rule as an advisory screen, and all four batch-4 attenuators pass the gauge (0.98 to 1.06). The nine second-print specimens joined the surrogate's training set as distinct observations at identical design coordinates (Section 3.4). None of the third-print specimens enters the fits, the cross-validation, or Fig. 6. Their sessions (2026-09-28 to 2026-10-02) postdate the most recent refit.

4.3 Surrogate Audits Across the Campaign. We audit each refit with leave-one-out cross-validation (LOOCV), using one full NUTS refit per held-out specimen, and with length-scale diagnostics. The seed-state SAASBO length-scale diagnostics (Fig. S2 of the Supplementary Information) show no input separating decisively from an equal share of the model sensitivity. At seven observations for six correlated inputs, these are prior-sensitive model diagnostics only, and they support no claim of variable relevance or ranking.

Held-out point accuracy *declined* as the campaign grew. The t_{180} LOOCV MAPE was 2.9% with rank correlation +0.76 on the seed state (8 specimens, 6 model inputs), 5.4% and +0.60 after batch 2 (17 specimens), and 6.2% and +0.19 after batch 3 (26 specimens, 12 inputs). The MAPE of the rebound objective we fit the model on, E_{reb}, was 29.3%, 23.6%, and 32.2% at the same states. We archive the fold-level records with the campaign data. Three changes coincided and are confounded: the input space doubled from 6 to 12 while the data grew only from 17 to 26, four of the six added axes carry a single value per batch and are therefore perfectly confounded with batch, and batch 3 itself added the campaign's largest outliers. The held-out posterior shrinks predictions toward the pooled mean, so it misses the campaign's extremes. We therefore read the predicted columns as a record of what the model expected when it chose each design, rather than as forecasts. Batch 4's value was the design family it selected and

Table 3 Model-recommended batch results, one row per Ax trial in ascending order, under the protocol and reporting conventions of Table 2 (CFC-180, stabilized means ± one standard deviation). The italic line above each batch gives its mass basis and the filament settings of its print job, read from the archived slicer project for batch 2 and from the design record for batches 3 and 4. Columns R to d_t and the two infill densities are the as-printed values after the constant-mass projection, and batch 2 did not vary infill. Each specimen received 18 to 20 stabilized drops against the seed batch's 99, and the archived per-drop records give the exact per-session counts. m is the weighed specimen mass, and I_{reb}, the mass-weighted rebound index of Eq. (1) (lower is better), uses it. "Pred." is the posterior-mean t_{180} archived when the model recommended the batch, before we tested any of its specimens. † marks a transmissibility-drift-flagged session whose mean is provisional, and ‡ marks an erratic session (per-drop CV above 1.5% without a drift flag). The print key (Table S4 of the Supplementary Information) maps each trial to its specimen. We generate the table from the archived data with `scripts/figures/build_batch_results_table.py`.

Trial	R	H	θ	d_s	d_t	Infill (%)		m	t_{180}	I_{reb}	Pred.
	(mm)	(mm)	(deg)	(mm)	(mm)	PLA	TPU	(g)	mean ± sd	(mg)	t_{180}
Batch 2, constant solid mass, infill not varied. PLA at 220 °C and 30 mm³/s, TPU at 240 °C and 2.5 mm³/s.											
10	21.9	52.7	80.0	10.53	2.63	-	-	17.91	1.0530 ± 0.0035	8.5	0.873
11	30.1	45.2	80.0	9.04	4.14	-	-	19.33	1.1847 ± 0.0076	9.6	0.885
12†	22.5	53.9	40.0	10.78	2.70	-	-	17.96	1.0411 ± 0.0154	7.9	0.926
13	27.3	65.5	80.0	8.08	3.28	-	-	19.38	0.9942 ± 0.0021	8.2	0.948
14	20.4	49.0	40.0	9.79	4.49	-	-	19.11	1.0548 ± 0.0042	3.9	0.953
15	34.9	52.3	40.0	10.47	2.62	-	-	18.51	0.9484 ± 0.0008	35.8	0.905
16	20.1	48.2	80.0	9.63	4.42	-	-	18.63	1.3337 ± 0.0117	7.5	0.910
17	26.9	74.0	80.0	8.08	3.70	-	-	19.98	1.0704 ± 0.0116	14.0	0.955
18	25.1	60.1	80.0	6.01	5.51	-	-	23.47	1.1516 ± 0.0040	8.6	0.987
Batch 3, constant printed mass (20.23 g). PLA at 217 °C and 20.5 mm³/s, TPU at 244 °C and 3.6 mm³/s.											
28	23.2	55.8	40.0	5.58	5.11	21	29	19.88	1.0373 ± 0.0024	5.6	1.059
29	27.8	76.5	40.0	8.35	3.83	13	31	19.92	1.0344 ± 0.0017	18.6	1.032
30	23.0	55.2	40.0	5.52	5.06	31	18	19.34	1.0425 ± 0.0009	6.3	1.058
31	20.7	65.1	40.0	9.04	4.14	18	21	19.92	1.0389 ± 0.0012	4.0	1.043
32	29.9	60.6	40.0	8.10	4.11	24	13	20.06	1.2513 ± 0.0096	14.9	1.020
33	23.1	55.4	40.0	5.54	5.08	29	16	19.33	1.0345 ± 0.0006	32.0	1.056
34	21.7	52.1	40.0	8.81	4.78	16	34	19.24	1.0471 ± 0.0040	5.3	1.065
35†	25.3	60.8	61.5	9.02	3.04	26	24	19.32	0.9804 ± 0.0051	29.2	1.039
36	25.9	62.1	40.0	8.27	3.11	34	26	19.31	1.0326 ± 0.0016	75.9	1.051
Batch 4, constant printed mass (20.23 g). PLA at 226 °C and 29.5 mm³/s, TPU at 236 °C and 2.6 mm³/s.											
37	34.3	67.1	79.0	8.98	2.58	22	21	19.62	0.8031 ± 0.0025	20.0	0.934
38	18.3	80.3	40.0	7.98	4.02	18	34	20.21	1.0846 ± 0.0030	18.8	1.008
39	34.7	68.3	80.0	8.86	2.60	21	34	19.64	0.9535 ± 0.0025	70.1	0.954
40†	20.8	91.6	40.0	5.00	4.58	18	21	20.58	1.0876 ± 0.0099	15.6	1.012
41‡	33.4	67.3	80.0	8.65	2.51	29	18	19.45	0.9106 ± 0.0181	7.5	0.943
42	17.7	77.7	40.0	8.47	3.88	19	34	20.20	1.0459 ± 0.0029	16.6	1.010
43	37.1	65.5	80.0	9.25	2.78	12	29	19.46	0.8766 ± 0.0036	75.3	0.942
44	20.8	91.5	40.0	4.99	4.58	19	21	20.56	1.0475 ± 0.0025	4.1	1.013
45	19.7	86.5	40.0	6.89	4.32	12	35	20.15	1.0705 ± 0.0013	17.0	1.004

the record it found, not its point accuracy, and its best-predicted specimen was the record.

Plain LOOCV also leaks once replicates exist. When we hold out one print of a repeated design, its twin remains in training at nearly identical coordinates, so the fold tests interpolation rather than prediction. We re-audited the full record with leave-one-design-out folds (35 folds over 44 specimens, each drran/2dran pair held out together, one NUTS refit per fold at the full budget of Section 3.4). The t_{180} MAPE is 4.8% with Spearman rank correlation $\rho_s = +0.32$ (two-sided permutation $p = 0.036$). While the measurements span 0.80 to 1.33, the held-out predictions stay near the pooled mean, with about a third of the measured spread (Fig. S3 of the Supplementary Information). The E_{reb} MAPE is 38.8% with $\rho_s = -0.40$ ($p = 0.008$), so the held-out ordering of E_{reb} runs against the measured one. Held-out design-level t_{180} skill is therefore weak but above chance. An earlier run of the same folds at a quarter of that NUTS budget gave $\rho_s = +0.08$ for t_{180} (we archive both runs), so this ranking statistic depends on how fully NUTS samples each fold's posterior, and we quote the full-budget run. Part of the apparent E_{reb} skill of the earlier states came from the model fitting chance differences between prints, which matches the poor print-to-print repeatability of the rebound objective in Fig. 9. The comparison to the LOOCV trajectory is descriptive rather than a controlled ablation. The states differ in specimen count, fold structure, and the presence of the repeat data itself.

The correlation between printed mass and transmissibility fell from $r = 0.82$ over the eight seed specimens to 0.12 over all 53 tested specimens ($r^2 = 0.01$), which suggests that the constant-printed-mass projection removed the seed batch's main confound. Within batch 4 alone the two correlate strongly ($r = 0.85$). There, however, the wide, high-twist family printed at 19.45 to 19.64 g and the tall, low-twist family at 20.15 to 20.58 g, so printed mass follows the design family. We therefore cannot read the correlation within the batch as a mass effect. We have not yet run the planned head-to-head calibration comparison against a point-estimated single-task GP with the same kernel, so we make no claim that SAASBO outperforms it here.

Studies pending. The following remain future work, and none of their values appear in this section. The batch-4 attenuators (the record specimen first) and the third-print outlier dran35 await reseats. The third print of the batch-3 designs has not yet entered the surrogate fit or its cross-validation. We generated a fifth batch, regenerated it once the second print entered the fit, and regenerated it again under I_{reb}, but did not print it, and nothing from it appears here. We have printed the 27 specimens of the replicate rank-stability study (Section 3.6), but we withhold their record until the repeats are complete. The quasi-static crush campaign and the metal-analog comparison (Supplementary Information, Section S11) have not yet run.

4.4 Simulated Campaign. Table 4 scores each run of the simulation-only campaign (Section 3.5) by its final hypervolume as a percentage of a best-known reference. That reference is the hypervolume of a 16,384-design reference sweep, a locally refined

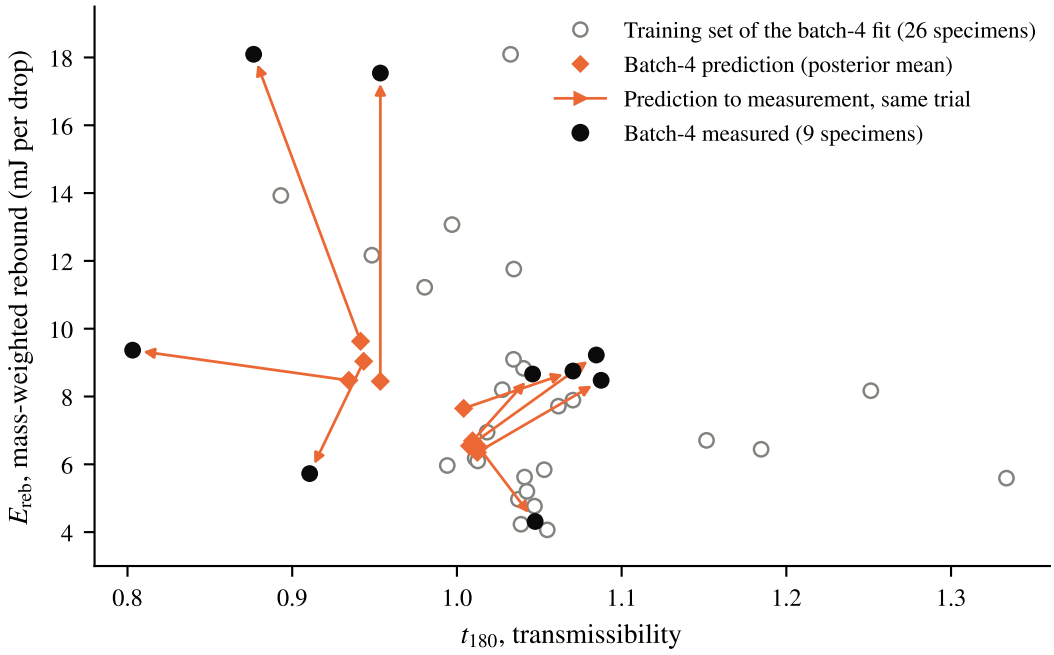


Fig. 8 Batch-4 predictions against outcomes. Orange diamonds are the archived posterior-mean predictions for trials 37 to 45, saved when the optimizer recommended the batch, and each arrow joins a prediction to the measured value of the specimen printed for the same trial (black). Gray circles are the 26 specimens we trained the batch-4 fit on (the seed batch, batch 2, and the first prints of batch 3). The vertical axis is E_{reb}, the first-power form of Section 3.3 that the optimizer predicted and minimized when it recommended the batch. Fig. 6 plots I_{reb} instead. We omit predictive uncertainty for legibility. The record attenuator corny7 (measured 0.803) is the left-most point. Lower is better on both axes.

front, and every feasible design that any compared run evaluated. After 45 simulated designs, BO reached 93.0% of this reference (seed range 90.0 to 96.1%), against 76.5 to 79.1% for the four baselines. BO finished ahead of every baseline on all ten seeds, and no baseline seed reached the worst BO seed, so the one-sided exact Mann-Whitney p is 5.4×10^{-6}, the smallest value ten runs against ten allow. The running hypervolume (Fig. S4 of the Supplementary Information) shows BO trailing the baselines during its Sobol batch and pulling ahead once the model chooses the designs.

We chose this objective pair before running the campaign, by screening six candidate pairs for the one whose front a random 45-design batch covers least (Section S14). The result therefore shows that the optimizer finds a front that sampling misses when one exists, not that it will on every pair. The control pair, t_{180} with peak tendon strain, shows the other case. There a random batch already collects about 85% of the pair's reference hypervolume, BO reached 82.6% against 77.4 to 79.8%, and no comparison survives a four-way Bonferroni correction (smallest exact $p = 0.018$). The control's baselines also used an earlier protocol that charged unprintable draws, so the two pairs are not a controlled contrast. Because the simulated struts are rigid, the simulator keeps t_{180} below about 1 and does not reproduce the amplification measured on hardware (Section 5). The simulated campaign therefore tests the optimizer only.

Table 4 Simulation-only campaign: final hypervolume after 45 designs as a percentage of each objective pair's best-known reference (mean ± sample standard deviation over ten seeds). p is the one-sided exact Mann-Whitney test that BO exceeds the baseline, and Wins counts the seeds on which BO finished ahead of the baseline run with the same seed index. LHS is Latin hypercube sampling. The envelope-pair baselines skip unprintable candidates without simulating them, and the strain control used the earlier protocol that charged them. We generate the table with `scripts/figures/sim_campaign_tables.py` from the per-seed records.

	t_{180} + envelope			t_{180} + strain	
Method	HV (%)	p	Wins	HV (%)	p
BO (qNEHVI)	93.0 ± 2.2			82.6 ± 4.3	
Random	79.1 ± 2.7	5.4×10^{-6}	10/10	79.6 ± 4.2	0.14
Sobol	77.3 ± 4.2	5.4×10^{-6}	10/10	79.8 ± 2.2	0.038
LHS	76.5 ± 3.4	5.4×10^{-6}	10/10	78.8 ± 2.8	0.018
Compass	78.1 ± 5.3	5.4×10^{-6}	10/10	77.4 ± 7.9	0.062

5 Discussion

What the campaign establishes. Across the four measured batches, the end-to-end loop closed repeatedly: printed specimens, instrumented characterizations, a fitted fully Bayesian surrogate, and three successive model-recommended batches whose own measurements fed the next refit. The measurement distinguishes the tested specimens. The campaign t_{180} means span 0.803 to 1.334, far beyond within-session drop scatter. The three prints of batch 3 locate the resolution floor for a single specimen. Clean seatings of different prints differ by a standard deviation of about 0.009, and a broadband gauge flags occasional seating failures of 10 to 20%. The physics is less favorable than the cushioning intuition suggests. At this scale and rate, most printed prisms *amplify* the peak shock reaching their top vertex (26 of 35 specimens measured above unity). We therefore sharpened the campaign question from "how much does the architecture cushion" to "which corner of the design space crosses into attenuation, and at what cost in rebound." The batches answered that question, though not on the first try. Batch 2's forecasts were optimistic on nine of nine specimens, and batch 3's best-predicted design measured as its worst amplifier on its first print, a seating artifact that its two later prints did not repeat. Batch 4 then delivered four attenuators from the wide, high-twist, thin-cable family, with its best-predicted design leading at $t_{180} = 0.803$. The strongest attenuators still tend to have low velocity loss indices (0.938 and 0.940 for corny9 and corny8),

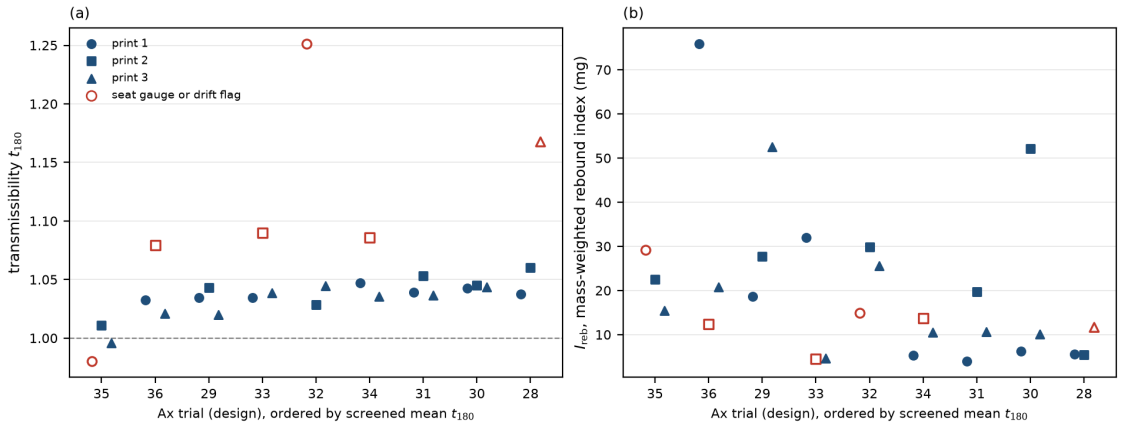


Fig. 9 Three prints of each batch-3 design. *(a)* Transmissibility and *(b)* I_{reb} for print 1 (circles), print 2 (squares), and print 3 (triangles), with the designs ordered by their screened mean transmissibility. Open red markers are sessions with a failed seat gauge or a drift flag, which the screened means leave out. The dashed line marks $t_{180} = 1$.

and the record specimen is mid-pack at 0.968. The tall batch-4 family reaches an index as high as 0.986 at a transmissibility near unity, so the front remains two-dimensional.

Pre-registered review of the objective choices. We pre-registered an adversarial, data-first review of the objective choices and ran it with the analysis tool named in the Acknowledgment. The review had to commit to its own objective set from the raw campaign files before reading ours. Its findings temper the round-1 formulation and set the round-3 revision. First, the rebound fraction e_{reb} rests on a ballistic interpretation of the second contact event that we have not yet validated against synchronized video (we have scheduled a restrained-versus-unrestrained check). Second, the apparent t_{180}-versus-E_{reb} trade-off (unchanged under I_{reb}, which orders the seed specimens identically) depends substantially on the single strongest attenuator (Spearman $\rho = -0.39$, $p = 0.38$ across the reliable specimens). Third, per-drop standard errors are much smaller than the between-specimen variation, and therefore understate design-level uncertainty. In response, we printed and tested the batch-3 plate three times, which measured the print-to-print scatter directly (Section 4.2) and confirmed the understatement at an order of magnitude in t_{180}. The between-seat replication rule now gates every extreme, the rebound fraction stays a candidate constraint pending its ballistic validation, and the constant-printed-mass projection removed the mass channel the review flagged. We report this trajectory because the loop is the contribution, and revising objectives mid-campaign, with the revision criteria stated in advance, is probably how such loops behave in practice.

Mass as a confound and as a design variable. Holding solid CAD mass constant did not hold printed grams constant. PLA prints at partial infill while thin TPU cables print near solid, so the seed specimens span 18.50 to 22.29 g, and among the mapped seed specimens mass and t_{180} correlate ($r = 0.82$). The geometry induces printed mass, which is strongly collinear with member dimensions (light specimens *are* the thick-strut, thin-cable corner by construction), so the seed observations cannot identify separate geometry and mass effects. A post hoc mass-adjusted regression would be unstable and would estimate a different, direct-effect quantity. We addressed the confound by experimental design rather than adjustment. We used the calibrated printed-mass model (Supplementary Information) to compare geometries at fixed printed grams in batches 3 and 4, where the batch mass CVs were 1.7 and 2.3%. Over all 53 tested specimens, mass and transmissibility are nearly uncorrelated ($r = 0.12$). We broke the confound by construction, and I_{reb} keeps grams penalized where they matter physically, in the motion returned to the payload.

Relation to prior art, and why simulation sits outside the loop. Relative to Pajunen et al. [11], we vary the geometry under an optimization loop rather than characterizing a single fabrication condition. Relative to Mo et al. [17], the loop's ground truth is physical measurement, and no simulated number enters the objectives, the acquisition, or candidate priority. We based that exclusion on an exploratory multi-engine simulation study that we conducted alongside the campaign (rigid-body through finite-element fidelities, archived in the project repository). The study could not reproduce the amplification that dominates the measured record, predicted design-invariant rebound where the measurement discriminates strongly, and correlated only weakly with the measured transmissibility. Its rigid-strut models cap the transmissibility near unity, while most printed specimens exceed it. Its constant-mass fairness constraints and its diagnosis of the seed batch's mass confound did shape the campaign design, which is the screening role we consider simulation to have earned here. Before simulation takes on a larger role, it must demonstrate prospective predictive skill on specimens it has not seen. The measured near-unity transmission of most seed specimens also cautions against reading the printed-tensegrity literature's attenuation results as generic. The as-printed cells here are not pretensioned, and activating genuine pretension in additively manufactured tensegrity (for example through shrink-on-treatment routes) is an open direction we deliberately scope out of the present campaign.

Practical considerations and path forward. PLA/TPU interface durability under repeated impact remains open. No joint failed during the repeated-drop measurement sessions, which covered more than 1,400 stabilized captures across 35 specimens, plus the second and third prints of batch 3. Because we did not design those sessions as controlled fatigue tests, the observation does not establish interface durability. Pull-out and fatigue characterization of the internal-anchor joint is planned follow-on work. Sensitivity to print-process covariates (the print key logs defects, humidity, and the mid-batch nozzle change for this purpose) and transfer to other material pairs likewise remain open. In the future we plan the re-seat confirmations of the batch-4 attenuators, the remaining replicate drop tests (Section 3.6), the metal-analog rank-transfer comparison (Supplementary Information, Section S11), quasi-static testing as an extension of the characterization (Section 3.3), pretension activation, tiled multi-cell absorbers and lattices [31], and miniaturization to crutch-tip envelopes. Long-term crutch use is associated with substantial upper-extremity overuse injury and ground-reaction-force impulses [57–

61]. Existing spring-loaded [62, 63] and polymer-damped [64] tip designs reduce loading rate but not peak transmitted force. Recent tensegrity crutch concepts [65] and open-source 3D-printed crutch work [66] suggest a plausible translation path, with adjacent footwear and orthotic lattice studies [67–69] providing transferable design heuristics.

Limitations. The measured evidence base is 53 specimens: 35 across the four batches and 18 re-prints of the nine batch-3 designs, which are the only designs we tested more than once. The surrogate fit includes the nine second prints but not yet the nine third prints. That replication exposed the sensitivity to seating and printing, so every single-specimen extreme, including the batch-4 record, is provisional until its re-seat. The drop protocol probes one height, one impact surface, and axial loading only. The specimens never leave the elastic regime, so crush-energy metrics await the quasi-static campaign. Per-specimen energy absorption in joules is also not recoverable from the current rig, because we do not instrument the falling-mass and displacement channels required for a force integral. The four filament axes carry one value per batch, so their effects are not identifiable. The held-out prediction skill of the surrogate declined as the input space grew, and over the full record it ranks held-out designs only weakly on t_{180} ($\rho_s = +0.32$) and inversely on E_{reb} ($\rho_s = -0.40$; Section 4.3). One tested seed specimen lacks a design mapping, one seed design's official specimen assignment carries a recorded discrepancy, and two seed designs have no session record. All results come from a single printer, material pair, and cell topology. We plan to address these limits with the replication rule, the re-seat confirmations, the completed replicate rank-stability study (Section 3.6), and the metal-analog protocol (Supplementary Information, Section S11).

6 Conclusions

We have presented an experiment-driven Bayesian-optimization framework for designing multi-material 3D-printed tensegrity-inspired energy absorbers, with TPU taking the place of the traditional tension elements and PLA members carrying compression most of the time. We have run it through the recommend, fabricate, measure, and update cycle three times on physical hardware. The framework parameterizes a T_3-prism family (five geometry variables, with infill and filament process variables joining them in batch 3) under a constant-mass fairness projection. It fabricates each candidate in a single multi-material FDM build with TPU tendons anchored inside the PLA strut ends and characterizes every specimen with repeated instrumented drops (99 for each seed specimen and 18 to 20 for later ones, after a warm-up). A fully Bayesian Gaussian-process surrogate [20, 22] fitted to those measurements recommends the next batch. Across the four measured batches (35 specimens), the transmissibility spanned 0.803 to 1.334, and most specimens amplified the peak shock. The model-recommended batches grew the dominated hypervolume 56% over the Sobol seed front, ending with a front whose five members are all model-chosen. The batch-4 best-predicted design set the program record at $t_{180} = 0.803$, 10 percent below the best seed specimen, and awaits its re-seat confirmation under the campaign's replication rule. Printing one batch's nine designs three times put the print-to-print standard deviation of the transmissibility at 0.009, an order of magnitude above the standard error of a session mean. A broadband seat gauge flagged both of the largest single-print errors, so a single-print extreme needs a replicate, or at least a clean re-seat, before it counts. [Placeholder: what the replicate study shows about rank stability.] The workflow is a candidate for other experimentally evaluated architected structures whose constitutive response is difficult to calibrate from first principles, but transfer beyond this printer, material pair, topology, and axial impact condition remains untested. In a simulation-only benchmark, BO collected 93% of the best-known hypervolume in 45 designs, where uninformed sampling collected 77 to 79%. On a control objective pair, where sampling alone covers most of the front, BO did not separate from the samplers.

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The authors used AI tools throughout this work, under their direction and with tightly coupled human-in-the-loop review and edits at every step, for text and figure generation, methodology, code, analysis, and testing. Claude Code (Anthropic) ran through the `anthropics/claude-code-action` GitHub Action from June to November 2026 and through its command-line interface from September to October 2026. Its models were Claude Opus 4.8, Claude Opus 5, Claude Opus 5.5, Claude Fable 5, and Claude Fable 5.1. GitHub Copilot ran as the Copilot coding agent from March to July 2026, mostly with Claude Opus 4.6, 4.7, and 4.8, and as the Copilot command-line interface and Copilot Autofix between March and June 2026. The Edison Scientific platform (data-analysis, literature, and precedent-search agents) produced mock peer reviews of earlier drafts, literature and reference checks, and the pre-registered review of the objective choices described in the Discussion (task `3e398131`, archived in the project repository) from May to September 2026. The faster-whisper speech-to-text model transcribed the co-authors' recorded reviews of the draft. The file `manuscript/AI-USE.md` in the project repository lists the tool, model, access route, and dates of each use. The human authors have reviewed the manuscript in full and take responsibility for it.

Funding Data

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Conflict of Interest

The authors declare no conflict of interest.

Data Availability Statement

The data and information that support the findings of this article are freely available at: https://github.com/vertical-cloud-lab/tensegrity-optimization. The repository holds the data snapshot behind every table and figure of this paper, with the scripts that regenerate them from it. It also holds the Bayesian-optimization campaign code, with the experiment state, recommendations, and archived predictions of each round, and the slicer project of each printed batch. The remaining contents are the parametric CAD model of the T_3 prism, the joint-design study, the drop-test processing pipeline with the per-drop records of every session, the cross-validation fold records, and the exploratory simulation study. We will mint a Zenodo DOI at acceptance.

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List of Figures

1 Closed-loop, experiment-driven design framework. Initialize and Conclude run once, and the dashed arrow returns each tested batch to the Design step. *Initialize:* the five-variable T_3-prism design space of Table 1 (Fig. 2). *Design:* a Gaussian-process surrogate fitted to the noisy measurements recommends the next batch (Section 3.4). *Make:* we scale each batch to a common mass and lay it out as one build plate. We slice the plate with PLA supports painted by hand along the TPU tendons and print it in one dual-nozzle job, with the hard material (PLA) from the left nozzle and the soft material (TPU) from the right (Section 3.2). *Test:* we drop-test the labeled specimens one at a time on the drop-tower carriage (Section 3.3). *Conclude:* report the final Pareto front. The plate render is batch 4, the mid-print frame is a later replicate plate, and the labeled specimens are batch 2. 3

2 Design space of the T_3-prism campaign. *(a)* A mid-range design of the five-variable box ($R = 32.5$ mm, $H = 85$ mm, $\theta = 60°$, $d_s = 9$ mm, $d_t = 4.25$ mm, before scaling to the mass target), labeled with the bounds of each variable. The side elevation is an orthographic projection, so each end triangle appears as a line, and H is the vertical distance between the planes through the joint centers. The plan view shows the base triangle (solid), the top triangle (dashed), the circumradius R, and the twist θ between them. *(b)* Six tested specimens at their printed, post-projection geometry and one millimeter scale, labeled with the batch, the Ax trial (or the Sobol spec of a seed specimen), and the measured t_{180} and E_{reb}. *(c)* Specimen corny7 as printed, with its ID label around one strut. The drawings show PLA struts in black, TPU tendons in green, and the square strut-end housings. We printed specimen corny7 in white PLA. 5

3 Three frames from the build-chamber camera timelapse of replicate plate 1 on the Bambu Lab H2D (https://youtu.be/EmCLms7-Bd8). The nine specimens print upright, with the prism axis normal to the build plate, beside a prime tower. (a) First layers, with the bottom TPU tendons and the strut-end housings. (b) Mid-height, with the struts and the painted PLA supports partly built. (c) Final layers, with the top tendons complete. We only crop and down-scale the frames. . . 6

4 Base-plate input and top-vertex response for one commissioning drop, shown to illustrate the transmissibility definition. The specimen is 57vqhx from the pre-campaign print-defect study (2026-07-28, event 50 of its session), dropped from 60 in onto the earlier stack of four felt pads and one cardboard sheet. Campaign drops land on the polyurethane surface instead. We low-pass filter both channels with the SAE J211/1 CFC filter, and the top-vertex trace is the magnitude of the three filtered axes. The zero-phase CFC-180 filter spreads the sub-millisecond pulse in both directions in time, so its traces are already nonzero at impact onset. (a) At CFC-180, the reporting class, the peak ratio gives $t_{180} = 588/563 = 1.045$. (b) At CFC-1000, the diagnostic class, the same drop gives $2394/1861 = 1.287$. 7

5 Specimens of the three replicate-study designs, as photographed for the print log. Rows are trials 37 (corny7), 39 (corny8), and 38 (corny2). The first column is the batch-4 specimen (print log of 2026-09-11), and the other three are the front-row copy from replicate plates 1 to 3 (print log of 2026-10-17). Labels give the specimen ID and weighed mass. The batch-4 struts are white PLA and the replicate struts black PLA. Camera distance varies between photographs, so apparent size is not comparable across cells. The replicate drop tests are not complete, so the figure reports no measured values. 9

6 Measured objective space and front evolution across the four batches. The vertical axis is the mass-weighted rebound index $I_{\mathrm{reb}} = m\, e_{\mathrm{reb}}^2$ of Eq. (1). Gray circles are all 35 measured specimens at their stabilized means. The blue lines are the non-dominated fronts after each batch, with the dominated hypervolume (HV) at the $(1.35, 50\,\mathrm{mg})$ reference in the legend. Batch 2 grew the front on the I_{reb} axis, batch 3 filled the knee, and batch 4 extended it to lower transmissibility. We draw repeated batch-3 designs at their first-print values. The text gives the design-mean front. Lower is better on both axes. 10

7 The nine Sobol seed designs at their printed geometry, drawn to one millimeter scale as in Fig. 2(b). We label each drawing with its Sobol spec and the ID of its printed specimen. We printed specs 03 and 06 (asterisks) but did not drop-test them, and we label spec 08 with its tested specimen, bag26v. 11

8 Batch-4 predictions against outcomes. Orange diamonds are the archived posterior-mean predictions for trials 37 to 45, saved when the optimizer recommended the batch, and each arrow joins a prediction to the measured value of the specimen printed for the same trial (black). Gray circles are the 26 specimens we trained the batch-4 fit on (the seed batch, batch 2, and the first prints of batch 3). The vertical axis is E_{reb}, the first-power form of Section 3.3 that the optimizer predicted and minimized when it recommended the batch. Fig. 6 plots I_{reb} instead. We omit predictive uncertainty for legibility. The record attenuator corny7 (measured 0.803) is the left-most point. Lower is better on both axes. 13

9 Three prints of each batch-3 design. *(a)* Transmissibility and *(b)* I_{reb} for print 1 (circles), print 2 (squares), and print 3 (triangles), with the designs ordered by their screened mean transmissibility. Open red markers are sessions with a failed seat gauge or a drift flag, which the screened means leave out. The dashed line marks $t_{180} = 1$. 14

List of Tables

1 Design variables for the T_3-prism campaign, as the BO code implements them. The seed batch is an $n = 9$ Sobol draw (fixed seed) over the five geometry variables, and we then project each design onto the constant-mass budget of the text. The process variables joined the space in batch 3. The acquisition chooses the two infill densities per specimen. The four filament settings admit one value per print job, and we therefore draw them as a single quasi-random point per batch, confounded with batch until a round varies them within a plate. We hold the joint diameter fixed. All specimens print upright on a Bambu Lab H2D, with the prism axis normal to the build plate and the base triangle on the bed (Fig. 1, Make panel). The seed batch used one specimen per named build plate, and each later batch used one nine-specimen plate. 4

2 Seed-batch drop-tower results: 99 stabilized drops per specimen (101 for autv5r) at 60 in onto the standardized polyurethane surface, CFC-180 filtering, counted after we discarded the first 2 warm-up drops of each session. Means ± one standard deviation over the stabilized drops. We order the rows by t_{180}. I_{reb} uses each specimen's weighed mass per Eq. (1). A dash denotes a quantity that we cannot compute from the available record. We exclude specimen *amd-jwm*, which lacks a weighed mass and a confirmed design mapping, from the surrogate fit and from all mass-dependent quantities. The ringdown frequencies of these sessions are in the Supplementary Information. We printed specs 03 and 06, but they have no drop sessions in the campaign record. We printed all specimens upright (Fig. 3). 9

3 Model-recommended batch results, one row per Ax trial in ascending order, under the protocol and reporting conventions of Table 2 (CFC-180, stabilized means ± one standard deviation). The italic line above each batch gives its mass basis and the filament settings of its print job, read from the archived slicer project for batch 2 and from the design record for batches 3 and 4. Columns R to d_t and the two infill densities are the as-printed values after the constant-mass projection, and batch 2 did not vary infill. Each specimen received 18 to 20 stabilized drops against the seed batch's 99, and the archived per-drop records give the exact per-session counts. m is the weighed specimen mass, and I_{reb}, the